

Metabolic and neurobehavioral disturbances induced by purine recycling deficiency in *Drosophila*

Reviewed Preprint

Revised by authors after peer review.

About eLife's process

Reviewed preprint version 2
February 21, 2024 (this version)

Reviewed preprint version 1
July 31, 2023

Posted to preprint server
June 23, 2023

Sent for peer review
June 3, 2023

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Abstract

Adenine phosphoribosyltransferase (APRT) and hypoxanthine-guanine phosphoribosyltransferase (HGPRT) are two structurally related enzymes involved in purine recycling in humans. Inherited mutations that suppress HGPRT activity are associated with Lesch-Nyhan disease (LND), a rare X-linked metabolic and neurological disorder in children, characterized by hyperuricemia, dystonia and compulsive self-injury. To date, no treatment is available for these neurological defects and no animal model recapitulates all symptoms of LND patients. Here we studied LND-related mechanisms in the fruit fly. By combining enzymatic assays and phylogenetic analysis, we confirm that no HGPRT activity is expressed in *Drosophila melanogaster*, making the APRT homologue (*Appt*) the only purine-recycling enzyme in this organism. Whereas APRT deficiency does not trigger neurological defects in humans, we observed that *Drosophila Appt* mutants show both metabolic and neurobehavioral disturbances, including increased uric acid levels, locomotor impairments, sleep alterations, seizure-like behavior, reduced lifespan, and reduction of adenosine signaling and content. Locomotor defects could be rescued by *Appt* re-expression in neurons and reproduced by knocking down *Appt* selectively in the protocerebral anterior medial (PAM) dopaminergic neurons, the mushroom bodies or glia subsets. Ingestion of allopurinol rescued uric acid levels in *Appt*-deficient mutants but not neurological defects, as is the case in LND patients, while feeding adenosine or *N*⁶-methyladenosine (*m*⁶A) during development fully rescued the epileptic behavior. Intriguingly, pan-neuronal expression of an LND-associated mutant form of human HGPRT (I42T), but not the wild-type enzyme, resulted in early locomotor defects and seizure in flies, similar to *Appt* deficiency. Overall, our results suggest that *Drosophila* could be used in different ways to better understand LND and seek a cure for this dramatic disease.

eLife assessment

The manuscript looks at how dysregulated purine metabolism in mutants for the *Aprt* gene impacts survival, motor and sleep behavior in the fruit fly. Interestingly, although several deficits arise from dopaminergic neurons, dopamine levels are increased in *Aprt* mutants. Instead the biochemical change responsible for *Aprt* mutant neurobehavioural phenotypes appears to be a reduction in levels of adenosine. This **valuable** study suggests that *Drosophila* *Aprt* mutants may serve as a model for understanding Lesch-Nyhan Disease (LND), caused by mutations in the human *HPRT1* gene, and may also potentially serve as a model to screen for drugs for the neurobehavioural deficits observed in LND. The strength of evidence is **solid**.

Introduction

The purine salvage pathway is an essential component of cellular metabolism that allows the recovery of free purine bases derived from the diet or from the degradation of nucleic acids and nucleotides, thus avoiding the energy cost of *de novo* purine biosynthesis (Nyhan et al. 2014). Energy-intensive tissues, such as cardiac muscle and brain cells, extensively use this pathway to maintain their purine levels (Ipata 2011 [↗](#), Johnson et al. 2019 [↗](#)). The two main recycling enzymes involved in the salvage pathway in mammals are hypoxanthine-guanine phosphoribosyltransferase (HGPRT), which converts hypoxanthine and guanine into IMP and GMP, respectively, and adenine phosphoribosyltransferase (APRT), which converts adenine into AMP.

APRT and HGPRT deficiencies induce very different disorders in humans. Loss of APRT seems to have only metabolic consequences, leading to the formation of 2,8-dihydroxyadenine crystals in kidney, which can be fatal but is readily prevented by allopurinol treatment (Bollée et al. 2012 [↗](#); Harambat et al. 2012 [↗](#)). In contrast, highly inactivating mutations in HGPRT trigger Lesch-Nyhan disease (LND), a rare neurometabolic X-linked recessive disorder with dramatic consequences for child neurodevelopment (Lesch and Nyhan 1964 [↗](#), Seegmiller et al. 1967 [↗](#)). The metabolic consequence of HGPRT deficiency is an overproduction of uric acid in the blood (hyperuricemia) that can lead to gout and tophi, or nephrolithiasis (Sass et al. 1965 [↗](#); Kelley et al. 1967 [↗](#)). Affected children also develop severe neurological impairments, such as dystonia, choreoathetosis, spasticity, and a dramatic compulsive self-injurious behavior (Nyhan 1997 [↗](#); Jinnah et al. 2006; Torres and Puig 2007 [↗](#); Schretlen et al. 2005 [↗](#); Madeo et al. 2019 [↗](#)). They have a developmental delay from 3 to 6 months after birth and most of them never walk or even sit without support. Xanthine oxidase inhibitors, such as febuxostat or allopurinol, are given to patients after diagnosis to decrease their uric acid levels and prevent the formation of urate crystals in kidney, which can lead to renal failure (Kelley et al. 1967 [↗](#); Torres et al. 2007 [↗](#); Lahaye et al. 2014 [↗](#)). However, there is as yet no treatment to alleviate the neurobehavioral symptoms of LND (Torres et al. 2007 [↗](#); Jinnah et al. 2010 [↗](#); Madeo et al. 2019 [↗](#)).

To date, the causes of neurobehavioral troubles in LND are still not elucidated (Jinnah et al. 2010 [↗](#), Bell et al. 2016 [↗](#)). The most favored hypothesis is a dysfunction of the brain's basal ganglia, and particularly of its dopaminergic pathways (Baumeister and Frye 1985 [↗](#); Visser 2000; Nyhan 2000 [↗](#); Saito and Takashima, 2000 [↗](#); Egami et al. 2007 [↗](#)). Indeed, analyses revealed a marked loss of dopamine (DA) (Lloyd et al. 1981 [↗](#); Ernst et al. 1996 [↗](#)) and DA transporters (Wong et al. 1996 [↗](#)) in the basal ganglia of LND patients. DA deficits have also been reported in HGPRT knockout rodents, but without motor or neurobehavioral defects (Finger et al. 1988 [↗](#); Dunnett et al. 1989 [↗](#); Jinnah et al. 1993 [↗](#); Jinnah et al. 1994 [↗](#); Meek et al. 2016 [↗](#)). Recent studies reported

that HGPRT deficiency disrupts proliferation and migration of developing midbrain DA neurons in mouse embryos, arguing for a neurodevelopmental syndrome (Witteveen et al. 2022 [\[1\]](#)). This could result from ATP depletion and impaired energy metabolism (Bell et al. 2021 [\[2\]](#)) or an overactivation of *de novo* purine synthesis leading to the accumulation of potentially toxic intermediates of this pathway (López, 2008 [\[3\]](#); López et al. 2020 [\[4\]](#)). Pharmacological models have also been developed by injecting the neurotoxin 6-hydroxydopamine into neonatally rat brains, which induced a self-mutilation behavior in response to DA agonist administration in adulthood. However, these models are of limited utility as they do not reproduce the basic genetic impairment of LND (Breese et al. 1990 [\[5\]](#); Knapp et al. 2016; Bell et al. 2016 [\[6\]](#)).

New animal models are therefore needed to study LND pathogenesis and find efficient therapeutic molecules. The fruit fly *Drosophila melanogaster* presents many advantages for translational studies and drug discovery (Fernández-Hernández et al. 2016 [\[7\]](#), Perrimon et al. 2016 [\[8\]](#); Papanikolopoulou et al. 2019 [\[9\]](#)). Although the importance of this invertebrate model for studying rare human genetic diseases is now recognized (Oriol and Lasko 2018 [\[10\]](#)), a *Drosophila* LND model has not yet been developed to our knowledge. This is probably due to the fact that no HGPRT activity has been detected in this organism (Miller and Collins 1973 [\[11\]](#); Becker 1974a [\[12\]](#); Becker 1974b [\[13\]](#)). However, an ortholog of APRT is expressed in *Drosophila* (Johnson et al. 1987 [\[14\]](#)), encoded by the *Aprt* gene. It is therefore possible that part of the functions of human HGPRT, and in particular those essential for nervous system development and physiology, are endorsed by *Aprt* in *Drosophila*.

Here, we studied the effects of *Aprt* deficiency on purine metabolism, lifespan and various behaviors, including spontaneous and startle-induced locomotion, sleep and seizure-like bang-sensitivity. We show that *Aprt* is required during development and in adult stage for many aspects of *Drosophila* life, and that its activity is of particular importance in subpopulations of brain dopaminergic neurons and glial cells. Lack of *Aprt* appears to decrease adenosinergic signaling and induces both metabolic and neurobehavioral symptoms in flies, as is the case with HGPRT in humans. We also find that expression of an LND-associated mutant form of HGPRT, but not the wild-type enzyme, in *Drosophila* neurons, induced neurobehavioral impairments similar to those of *Aprt*-deficient flies. Such a potential toxic gain-of-function effect of mutated HGPRT had not yet been demonstrated in an animal model.

Results

Evolution of HGPRT proteins

As shown in Figure S1, the pathways of purine anabolism/catabolism and recycling have been closely conserved between *Drosophila* and humans: all genes related to purine metabolism have homologs in both species, except for the human *HPRT1* gene encoding HGPRT (step 13 in Figure S1) which has no orthologue in flies (see below), and the lack of urate oxidase in humans (step 20). In accordance with reports from about 50 years ago (Miller and Collins 1973 [\[11\]](#); Becker 1974a [\[12\]](#); 1974b [\[13\]](#)), we confirmed that no HGPRT enzymatic activity can be detected in extracts of wild-type *Drosophila melanogaster* (shown in Table S2), using either hypoxanthine or guanine as substrate in the reaction medium. This intriguing observation prompted us to carry out a more precise analysis of the evolution of HGPRT.

HGPRT proteins are ancient, for they are present in both Bacteria and Archaea. However, the analysis of the phyletic distribution of HGPRT proteins revealed their striking rareness in Insecta. This conclusion is based on PSI-Blast sequence similarity searches on the NCBI Insecta database (taxid: 6960, 50557). The phylogenetic analysis showed that all the 11 HGPRT proteins found in Insecta cluster mainly with Bacteria, but also with Fungi, Apicomplexa, and Acari (see Figure S2, *red font*, see legend for details). These and further evidence support the acquisition of HGPRT in a

few Insecta species by horizontal gene transfer (G. Matassi, unpublished observations). In particular, HGPRT has no homologue in Drosophilidae, with the potential exception of a single species, *Drosophila immigrans*, in which our most recent PSI-BLAST analysis detected one hit (accession KAH8256851.1, annotated as hypothetical protein). Yet this sequence is 100% identical to the HGPRT in the Gammaproteobacterium *Serratia marcescens*. A phylogenetic analysis showed that the *D. immigrans* HGPRT clusters with the *Serratia* genus (Figure S3), suggesting either a contamination of the sequenced sample or a very recent horizontal gene transfer event. The second scenario is more likely for the corresponding nucleotide sequences differ by 5 synonymous substitutions (out of 534 positions). We also carried out structural similarity searches against the RCSB Protein Data Bank repository, using the human HGPRT structure as query (PDB identifiers: 5HIA or 1Z7G). This analysis did not identify any protein with a divergent sequence and relevant similarity with HGPRT 3D structure in *Drosophila melanogaster*, consistent with the lack of HGPRT enzymatic activity in this organism.

***Drosophila* lacking Aprt activity have a shortened lifespan**

Both the phylogeny and enzymatic assays therefore strongly suggest that Aprt (Figure S1, *step 12*) is the only recycling enzyme of the purine salvage pathway in *Drosophila*. To assess the importance of purine recycling in brain development and function in this organism, we analyzed the phenotypes induced by a deficiency in Aprt. The *Aprt*⁵ mutant was originally generated by chemical mutagenesis followed by selection of flies resistant to purine toxicity (Johnson and Friedman, 1981 [↗](#); Johnson and Friedman, 1983 [↗](#)). Enzymatic assays confirmed a strong reduction in Aprt activity in extracts of heterozygous *Aprt*^{5/+} mutants and its absence in homozygous and hemizygous (*Aprt*⁵/*Df(3L)ED4284*) mutants (Figure S4 and Table S1). Sequencing of the *Aprt*⁵ cDNA (Figure S5A) indicated that the *Aprt*⁵ mRNA codes for a protein with several amino acid changes compared to *Drosophila melanogaster* wild-type Aprt, modifying in particular 3 amino acid residues that have been conserved in the Aprt sequences from *Drosophila* to humans (Figure S5B). These mutations are likely to be responsible for the loss of enzymatic activity. The homozygous *Aprt*⁵ mutants are considered viable because they develop and reproduce normally. However, we observed that these mutants have a significantly reduced longevity, their median lifespan being only 38 days against 50 days for wild-type flies ($p < 0.001$) (Figure 1A [↗](#)).

Uric acid levels are increased in *Aprt*⁵ mutants and rescued by allopurinol

In humans, one of the consequences of HGPRT deficiency on metabolism is the overproduction of uric acid (Harkness et al. 1988 [↗](#); Fu et al. 2015 [↗](#)). We assayed the levels of purine metabolites by HPLC and found that the level of uric acid was substantially increased by 37.7% on average in *Drosophila Aprt*⁵ mutant heads ($p < 0.01$) (Figure 1B [↗](#)). We then tried to rescue uric acid content in flies by providing allopurinol in the diet, as is usually done for LND patients. Allopurinol is a hypoxanthine analog and a competitive inhibitor of xanthine oxidase, an enzyme that catalyzes the oxidation of xanthine into uric acid (Figure S1, *step 19*). Remarkably, the administration of 100 µg/ml allopurinol during 5 days decreased uric acid levels to a normal concentration range in *Aprt*⁵ mutant heads (Figure 1B [↗](#)). Therefore, quite similarly to HGPRT deficit in humans, the lack of Aprt activity in flies increases uric acid levels and this metabolic disturbance can be prevented by allopurinol.

Aprt deficiency decreases motricity in young flies

LND patients present dramatic motor disorders that prevent them for walking at an early age. To examine whether a deficiency in the purine salvage pathway can induce motor disturbance in young flies, we monitored the performance of *Aprt*-null mutants in startle-induced negative geotaxis (SING), a widely used paradigm to assess climbing performance and locomotor reactivity (Feany and Bender, 2000 [↗](#); Friggi-Grelin et al. 2003 [↗](#); Riemensperger et al. 2013 [↗](#); Sun et al.

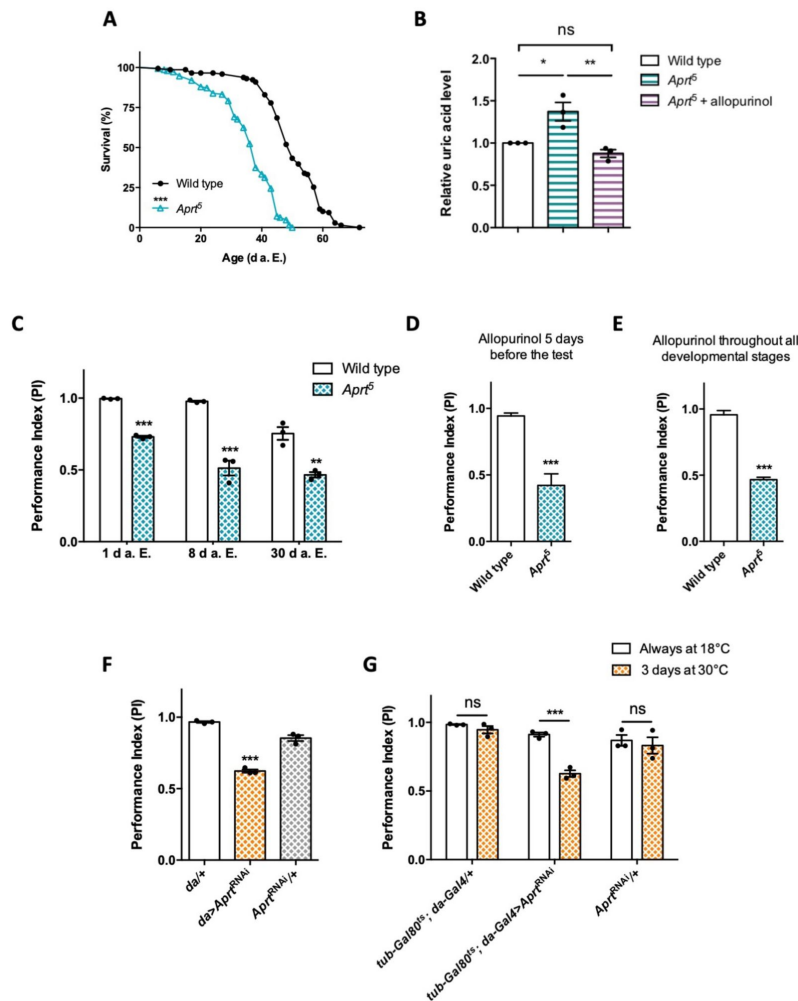


Figure 1.

Aprt deficiency shortens lifespan and induces metabolic and neurobehavioral defects.

(A) *Aprt*⁵ mutant flies have a reduced lifespan compared to wild-type flies (median lifespan: 38 and 50 days, respectively). Three independent experiments were performed on 150 males per genotype with similar results and a representative experiment is shown. Log-rank test (***p < 0.001). (B) HPLC profiles on head extracts revealed an increase in uric acid levels in *Aprt*⁵ mutant flies. Administration of 100 µg/ml allopurinol for 5 days before the test rescued uric acid levels. Mean of 3 independent experiments performed on 40 flies per genotype. One-way ANOVA with Tukey's *post-hoc* test for multiple comparisons (*p < 0.05; **p < 0.01; ns: not significant). (C-E) Effect on climbing ability. (C) *Aprt*⁵ mutants show early defects in the SING paradigm that monitors locomotor reactivity and climbing performance. This deficit was already obvious at 1 d.a.E. (days after eclosion) and further decreased up to 8 d.a.E., after which it did not change significantly up to 30 d.a.E. Mean of 3 independent experiments performed on 50 flies per genotype. Unpaired Student's *t* test (**p < 0.01; ***p < 0.001). (D, E) Administration of allopurinol does not rescue the motricity defects of *Aprt*-deficient mutants. Feeding the *Aprt*⁵ mutants with allopurinol (100 µg/ml) either in adults 5 days before the test (D) or throughout all developmental stages (E) did not alter the defects observed in SING behavior. Results of one experiment performed on 50 flies per genotype at 10 d.a.E. Unpaired Student's *t* test (***p < 0.001). (F) Downregulating *Aprt* by RNAi in all cells (*da>Aprt*^{RNAi}) also led to an early impairment in climbing responses in the SING assay at 10 d.a.E., as compared to the driver (*da/+*) and effector (*Aprt*^{RNAi/+}) only controls. Mean of 3 independent experiments performed on 50 flies per genotype. One-way ANOVA with Tukey's *post-hoc* test for multiple comparisons; (***p < 0.001). (G) Adult-specific inactivation of *Aprt* (*tub-Gal80^{TS}; da-Gal4>Aprt*^{RNAi}) decreased startle-induced climbing abilities in the SING paradigm, suggesting that the locomotor impairment induced by *Aprt* deficiency is not caused by a developmental effect. Flies were raised at permissive temperature (18°C) in which *Gal80^{TS}* suppressed *Gal4*-controlled *Aprt*^{RNAi} expression, and were shifted from 18 to 30°C for 3 days before the test (between 7 and 10 d.a.E.) to activate transgene expression. Mean of 3 independent experiments performed on 50 flies per genotype. Two-way ANOVA with Sidak's *post-hoc* test for multiple comparisons (***p < 0.001; ns: not significant).

2018 [\[1\]](#)). Strikingly, *Aprt*⁵ mutant flies showed a very early SING defect starting from 1 day after eclosion (d a.E.) (performance index (PI) = 0.73 vs. 0.98 for wild-type flies, $p < 0.001$) that was more pronounced at 8 d a.E. (PI = 0.51 vs. 0.96 for the wild-type flies, $p < 0.001$). The fly locomotor performance did not further decline afterwards until 30 d a.E. (**Figure 1C** [\[1\]](#)). *Df(3L)ED4284* and *Df(3L)BSC365* are two partially overlapping small genomic deficiencies that uncovers *Aprt* and several neighbor genes. Hemizygous *Aprt*⁵/*Df(3L)ED4284* or *Aprt*⁵/*Df(3L)BSC365* flies also displayed SING defects at 10 d a.E. (PI = 0.71 and 0.68, respectively, compared to 0.97 for wild-type flies, $p < 0.01$) (Figure S6). In contrast to its beneficial effect on uric acid levels, we observed that allopurinol treatment did not improve the locomotor ability of *Aprt*⁵ mutant flies, either administered 5 days before the test or throughout the development (**Figure 1D** [\[1\]](#), E). This is comparable to the lack of effect of this drug against neurological defects in LND patients.

To confirm the effect of *Aprt* deficiency on the SING behavior, we used an *UAS-*Aprt*^{RNAi}* line that reduced *Aprt* activity by more than 95% when expressed in all cells with the *da-Gal4* driver (Table S1). These *Aprt* knocked down flies also showed a strong SING defect at 10 d a.E. (PI = 0.62 against 0.97 and 0.85 for the driver and *UAS-*Aprt*^{RNAi}* alone controls, respectively, $p < 0.001$), like the *Aprt*⁵ mutant (**Figure 1F** [\[1\]](#)). Next, we used *tub-Gal80^{ts}*, which inhibits Gal4 activity at permissive temperature (McGuire et al. 2003 [\[2\]](#)), to prevent *Aprt* knockdown before the adult stage. *tub-Gal80^{ts}*; *da-Gal4>Aprt^{RNAi}* flies raised at permissive temperature (18°C) did not show any locomotor impairment (**Figure 1G** [\[1\]](#), white bars). However, after being transferred for 3 days (between 7 and 10 d a.E.) at a restrictive temperature (30°C) that inactivates Gal80, the same flies demonstrated a similar SING defect as *Aprt*⁵ mutants (PI = 0.63 compared to 0.94 and 0.83 for the driver and RNAi alone controls, respectively, $p < 0.001$) (**Figure 1G** [\[1\]](#), yellow bars with dots). This demonstrates that *Aprt* inactivation at the adult stage only is sufficient to alter SING performance in *Drosophila*, strongly suggesting that this genotype does not result from developmental defects.

Cell specificity of *Aprt* requirement for startle-induced locomotion

We then tried to identify the neural cells in which *Aprt* is required to ensure a normal locomotor reactivity in young flies by expressing *Aprt^{RNAi}* with selective Gal4 drivers. Expression in all neurons with *elav-Gal4* led to decreased locomotor performance in the SING test (PI = 0.68 at 10 d a.E., vs. 0.90 and 0.86 for the driver and RNAi controls, respectively, $p < 0.05$) (**Figure 2A** [\[1\]](#)), which was comparable to the effects observed with the *Aprt*⁵ mutant or after ubiquitous expression of the RNAi. To confirm the role of neuronal *Aprt* in locomotor control, we generated a *UAS-*Aprt** line which allowed for a substantial increase in *Aprt* expression and activity (Figure S7). We then found that re-expressing *Drosophila* *Aprt* selectively in neurons in *Aprt*⁵ background partially but significantly rescued the SING phenotype of the null mutant (PI = 0.70 vs. 0.52 for *driver* and *UAS-*Aprt** controls in *Aprt*⁵ background, $p < 0.05$) (**Figure 2B** [\[1\]](#)).

Furthermore, *Aprt* knockdown in all glial cells with *repo-Gal4*, or in sub-populations of glial cells that express the glutamate transporter *Eaat1* with *Eaat1-Gal4*, which includes astrocyte-like glia, cortex glia and some subperineurial glia (Rival et al. 2004 [\[3\]](#); Mazaud et al. 2019 [\[4\]](#)), also led to a lower SING performance of 10-day-old flies (PI = 0.72 for *repo-Gal4* vs. 0.91 for both controls, $p < 0.05$, and PI = 0.56 for *Eaat1-Gal4* vs. 0.77, $p < 0.05$, and 0.88, $p < 0.01$, for the driver and RNAi controls, respectively) (**Figure 2C** [\[1\]](#), D). In contrast, *MZ0709-Gal4* (Doherty et al. 2009 [\[5\]](#)) and *NP6520-Gal4* (Awasaki et al. 2008 [\[6\]](#)) that selectively target the ensheathing glia, did not induce any significant locomotor defects when used to express the *Aprt* RNAi (Figure S8). Noticeably, re-expressing *Aprt* with *Eaat1-Gal4* in the *Aprt*⁵ background did not rescue the SING phenotype (**Figure 2E** [\[1\]](#)), in contrast to the positive effect of neuronal re-expression (**Figure 2B** [\[1\]](#)). Overall this suggests that *Aprt* is required both in neurons and glia subsets to ensure a normal SING performance in young flies, and that neuronal but not glial *Aprt* re-expression is sufficient to ensure a partially functional locomotor behavior.

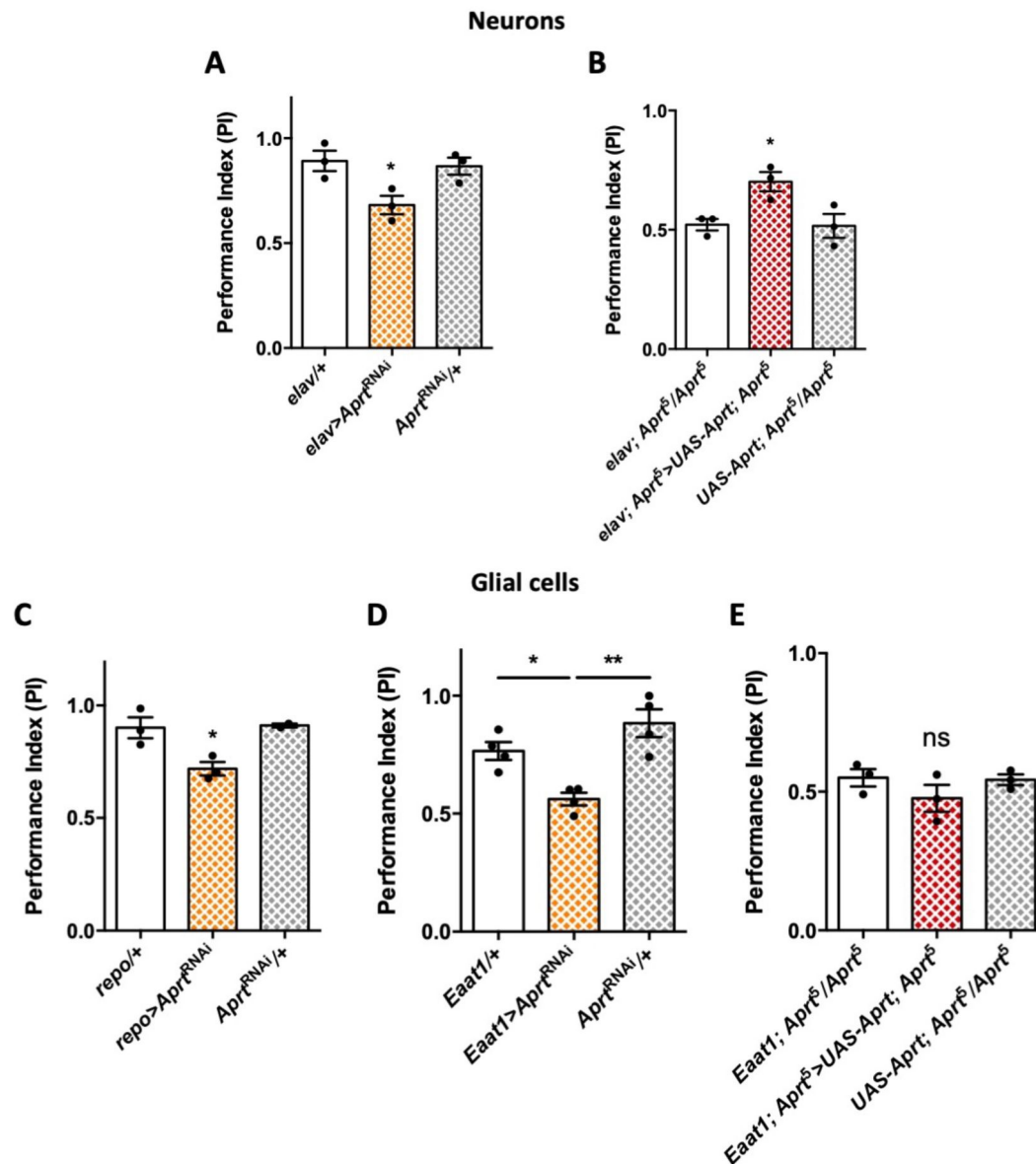


Figure 2.

***Apt* knockdown in neurons or glial cells disrupts startle-induced locomotion in *Drosophila*.**

(A) *Apt^{RNAi}* expression in all neurons with *elav-Gal4* decreased SING performance in 10-day-old flies. (B) Pan-neuronal expression of *Drosophila Apt* with the UAS-Gal4 system partially rescued the locomotor response of *Apt^Δ* mutants. (C-D) Downregulation of *Apt* expression in all glia with *repo-Gal4* (C) or in glial cells that express the glutamate transporter *Eaat1* with *Eaat1-Gal4* (D) also altered SING performances. (E) *Apt* re-expression in glial cells with *Eaat1-Gal4* driver did not rescue the climbing response of *Apt^Δ* mutants. Results of 3 or 4 independent experiments performed on 50 flies per genotype at 10 d a. E. One-way ANOVA with Tukey's *post-hoc* test for multiple comparisons (* $p < 0.05$; ** $p < 0.01$; ns: not significant).

To identify the neuronal subpopulations in which *Appt* is required to ensure proper locomotor responses in young flies, we first tested dopaminergic drivers since we previously showed that DA neurons play an important role in the control of the SING behavior (Riemensperger et al. 2011 [↗](#); Riemensperger et al. 2013 [↗](#); Sun et al. 2018 [↗](#)). *Appt* knockdown targeted to these neurons with *TH-Gal4* did not induce significant impairments (**Figure 3A** [↗](#)). This driver expresses in all brain DA neurons except a large part of the protocerebral anterior medial (PAM) cluster (Friggi-Grelin et al. 2003 [↗](#); Mao and Davis 2009; Pech et al. 2013 [↗](#)). In contrast, downregulating *Appt* with the *TH-Gal4*, *R58E02-Gal4* double driver, which labels all DA neurons including the PAM cluster (Sun et al. 2018 [↗](#)), induced a quite similar locomotor defect as did pan-neuronal *Appt* knockdown (PI = 0.72 vs. 0.96 and 0.93 for the driver and RNAi controls, respectively, $p < 0.001$) (**Figure 3B** [↗](#)). Besides, downregulating *Appt* in a majority of the serotonergic neurons with *TRH-Gal4* (Cassar et al. 2015 [↗](#)) did not induce a SING defect (**Figure 3C** [↗](#)).

These results suggest that some DA neurons in the PAM cluster are specifically involved in the locomotor impairments induced by *Appt* deficiency. It has been previously shown in our laboratory that inhibiting DA synthesis in a subset of 15 PAM DA neurons cluster was able to alter markedly SING performance in aging flies (Riemensperger et al. 2013 [↗](#)). We and others also reported that the degeneration or loss of PAM DA neurons is involved in the SING defects observed in several *Drosophila* models of Parkinson disease (Riemensperger et al. 2013 [↗](#); Bou Dib et al. 2014 [↗](#); Tas et al. 2018 [↗](#); Pütz et al. 2021 [↗](#)). Here we found that expressing *Appt*^{RNAi} in the PAM cluster with *R58E02-Gal4* reproduced the same motor disturbance as pan-neuronal expression (PI = 0.74 vs. 0.96, $p < 0.001$, and 0.85, $p < 0.05$, for the driver and RNAi controls, respectively) (**Figure 3D** [↗](#)), and this result was confirmed by using two other PAM drivers (*NP6510-Gal4* – expressing only in 15 neurons $\mp$ and *R76F05-Gal4*) (**Figure 3E** [↗](#), F). This strongly suggests that purine recycling deficiency compromises the correct functioning of these neurons, leading to a defective startle-induced locomotion.

Because PAM DA neurons innervate the horizontal lobes of the mushroom bodies (Liu et al. 2012 [↗](#); Riemensperger et al. 2013 [↗](#)), and because this structure has been shown to be involved in the control of startle-induced climbing (Riemensperger et al. 2013 [↗](#); Bou Dib et al. 2014 [↗](#); Sun et al. 2018 [↗](#)), we also inactivated *Appt* in MB neurons with *238Y-Gal4* that strongly expresses in all the MB cells (Aso et al. 2009 [↗](#)). Interestingly, this driver did induce a locomotor reactivity impairment (PI = 0.70 vs. 0.89 for both controls, $p < 0.01$) (**Figure 3G** [↗](#)), and the same result was observed with *VT30559-Gal4*, which is a very specific driver for all the MB intrinsic neurons (Plačaiš et al. 2017 [↗](#)) (**Figure 3H** [↗](#)). Overall, these results show that normal expression of the SING behavior depends on *Appt* expression in the PAM and MB neurons in *Drosophila*.

Sleep disturbances induced by *Appt* deficiency

Both the mushroom body and subpopulations of PAM DA neurons are known to be regulators of sleep in *Drosophila* (Nall et al. 2016 [↗](#); Artiushin and Sehgal 2017 [↗](#)). The fact that *Appt* deficiency in some of these cells impaired locomotor regulation prompted us to monitor the spontaneous locomotion and the sleep pattern of *Appt* mutants. Compared to controls, *Appt*-deficient flies did not have an altered circadian activity profile (Figure S9), nor any difference in total spontaneous locomotor activity during the day, as quantified by the number of infra-red beam cuts (*events*) per 30 min, but they showed a 26.2% increase in total activity during the night ($p < 0.001$) (**Figure 4A** [↗](#)). As usual, a sleep bout was defined as 5 min or more of fly immobility (Huber et al. 2004 [↗](#)) and we checked that wild-type and *Appt* mutant flies that did not move for 5 min were indeed asleep because they were less sensitive to mild mechanical stimulation (Figure S10). We found that *Appt*⁵ mutants slept significantly less than wild-type flies and that it was the case both during day and night (**Figure 4B** [↗](#), C). These mutants indeed showed a reduced walking speed (**Figure 4D** [↗](#)) and a smaller average sleep bout duration (**Figure 4E** [↗](#)), indicating a difficulty to maintain sleep. The reduced speed does not seem to be caused by a decreased energetic metabolism, since we could not detect different ATP levels in head and thorax of *Appt*⁵ mutants compared to wild-type flies (Figure S11).

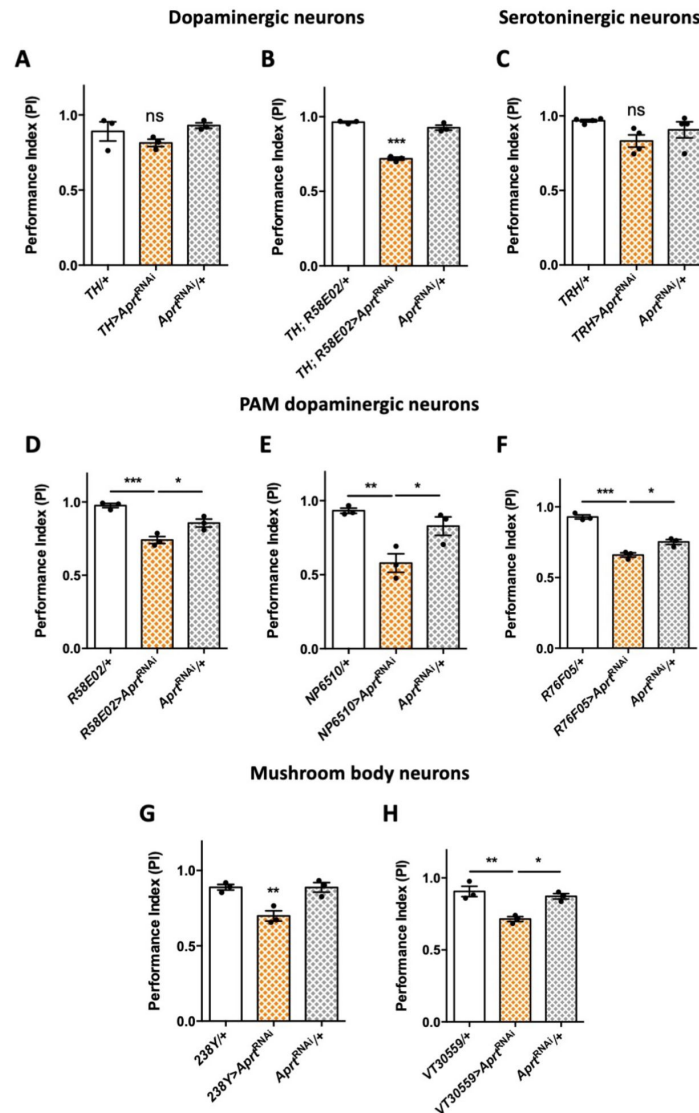


Figure 3.

***Aprt* downregulation in DA neurons of the PAM cluster and in mushroom body neurons impairs startle-induced locomotion.**

(A) RNAi-mediated *Aprt* inactivation in brain DA neurons except a large part of the PAM cluster with the *TH-Gal4* driver did not lead to locomotor defects in the SING assay. (B) In contrast, *Aprt* knockdown in all dopaminergic neurons including the PAM cluster with the *TH-Gal4*, *R58E02-Gal4* double-driver led to a decrease in SING performance. (C) *Aprt* downregulation in serotonergic neurons with *TRH-Gal4* did not alter startle-induced climbing response of the flies. (D-F) *Aprt* knockdown selectively in DA neurons of the PAM cluster using either *R58E02-Gal4* (D), *NP6510-Gal4* (E) or *R76F05-Gal4* (F) significantly decreased climbing performance. (G-H) *Aprt* knockdown in all the mushroom body intrinsic neurons with *238Y-Gal4* (G) or *VT30559-Gal4* (H) also led to a decrease in SING performance. Results of 3 or 4 independent experiments performed on 50 flies per genotype at 10 d a. E. One-way ANOVA with Tukey's *post-hoc* test for multiple comparisons (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; ns: not significant).

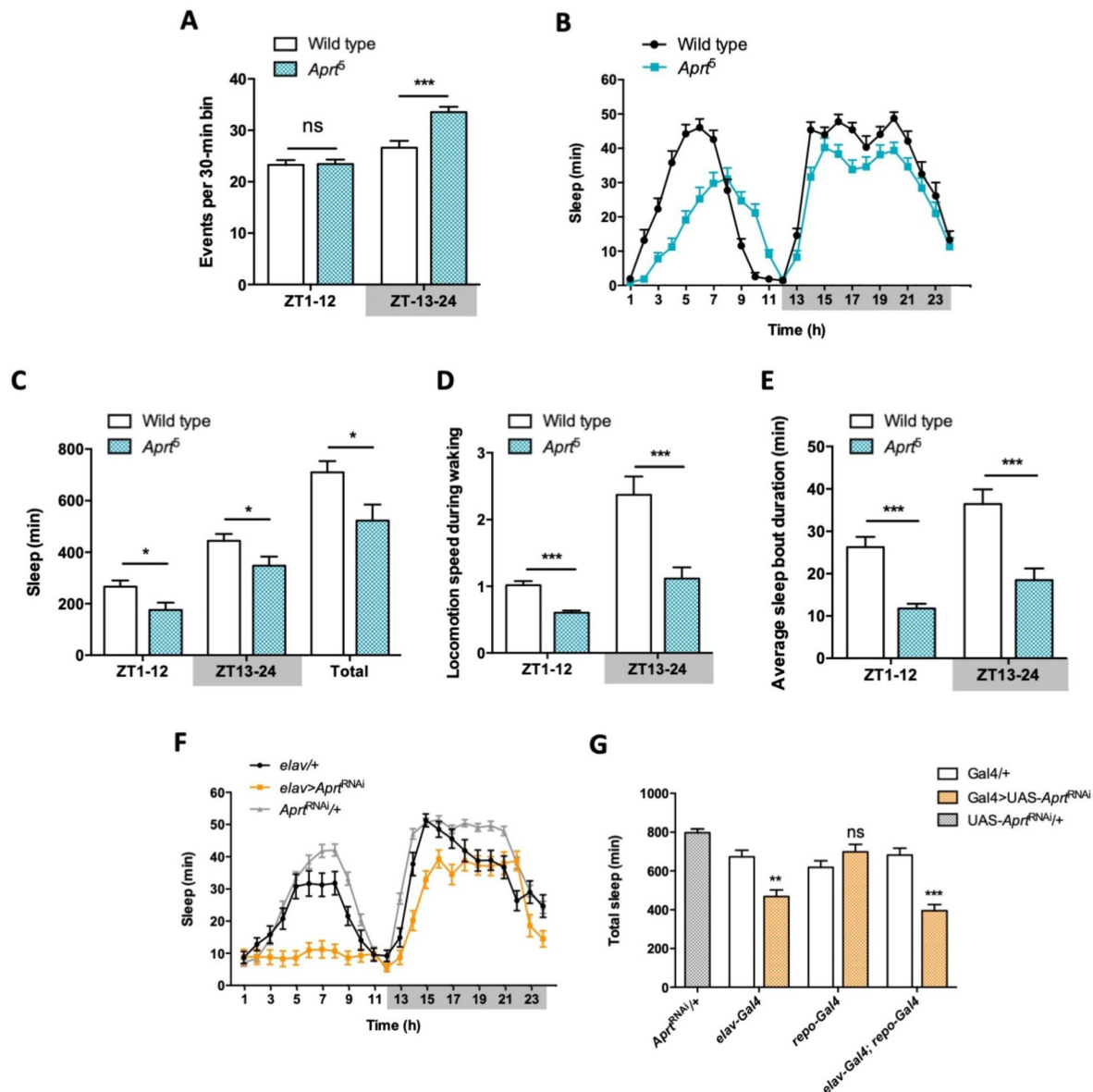


Figure 4.

Aprt-deficient flies sleep less and walk slower than wild-type flies.

(A) Quantification of total spontaneous locomotor activity during day and night over 5 LD cycles. *Aprt*⁵ mutants show no difference in spontaneous locomotion with wild-type flies during the day but a higher activity at night. 3 independent experiments were performed on 32 flies per genotype and mean $\pm$ SEM was plotted. Unpaired Student's *t* test (****p* < 0.001; ns: not significant). (B) Sleep pattern during a typical 24h LD cycle showing that the total amount of sleep is smaller during day and night in *Aprt*⁵ mutants compared to wild-type flies. (C) Quantification of day (ZT1-12), night (ZT13-24) and total sleep in *Aprt*⁵ mutants. ZT is for zeitgeber. (D) Locomotion speed during waking is reduced in *Aprt*⁵ mutants. (E) The average sleep bout duration is also decreased, indicating that *Aprt*⁵ mutants have a difficulty to maintain sleep. (F) Sleep pattern of *elav*>*Aprt*^{RNAi} flies, showing that knockdown of *Aprt* in all neurons led to sleep reduction during the night, similarly to the mutant condition, and an even more profound sleep defect during the day. (G) Quantification of total amount of sleep when *Aprt* was downregulated in all neurons (*elav*-Gal4), all glial cells (*repo*-Gal4) and both neurons and glial cells (*elav*-Gal4; *repo*-Gal4). Except for glia only, the resulting effect was a significant sleep reduction. For sleep and locomotion speed, means $\pm$ SEM were plotted. Unpaired Student's *t* test (C-E) and one-way ANOVA with Tukey's *post-hoc* test for multiple comparisons (G) (**p* < 0.05; ***p* < 0.01; ****p* < 0.001; ns: not significant).

Interestingly, RNAi-mediated downregulation of *Appt* selectively in neurons (*elav>Appt^{RNAi}* flies) also significantly decreased sleep duration during day and night (**Figure 4F** [↗](#)), whereas glial-only expression (*repo>Appt^{RNAi}*) had no effect (Figure S12A). Expressing the RNAi in both neurons and glia (*elav; repo>Appt^{RNAi}*) had similar effects as in neurons alone (Figure S12B). Quantification of total sleep in these experiments, and the total amount of day and night sleep, are shown in **Figure 4G** [↗](#) and Figure S12C, D, respectively. Overall, these results suggest that *Appt* expression is selectively needed in neurons for a normal sleep pattern in *Drosophila*.

Brain DA synthesis and levels are increased in *Appt*-deficient flies

Since we found that induced locomotion and sleep, two behaviors controlled by DA neurons in *Drosophila*, were altered in *Appt*-deficient flies, we logically expected brain DA levels to be reduced in these mutants, as is the case in the basal ganglia of LND patients. We therefore carried out comparative immunostaining for tyrosine hydroxylase (TH), the specific enzyme for DA synthesis (Friggi-Grelin et al. 2003 [↗](#); Riemensperger et al. 2011 [↗](#)), on dissected adult brains from wild-type and *Appt⁵* mutant flies. However, the global TH protein level appeared not to be decreased, but relatively increased by 17.5% in the *Appt⁵* mutant brain ($p < 0.01$), in particular around the mushroom bodies, a structure that receives dense dopaminergic projections (**Figure 5A** [↗](#), B). Moreover, DA immunostaining carried out on whole-mount dissected brains revealed a similarly increased level of this neuromodulator in *Appt⁵* flies, by 17.0% on average in the entire brain ($p < 0.01$), but not specifically in the mushroom bodies or another part of the brain (**Figure 5C** [↗](#), D). We also found that the transcript levels of *DTH1*, encoding the TH neuronal isoform in *Drosophila*, were increased in *Appt⁵* mutants compared to wild-type flies (**Figure 5E** [↗](#)), and, conversely, decreased when *Appt* was overexpressed ubiquitously (**Figure 5F** [↗](#)). Western blot experiments further indicated that DTH1 protein levels are increased in *Appt⁵* compared to controls (**Figure 5G-H** [↗](#)). This indicates that disruption of the purine salvage pathway does not impede DA synthesis and levels in the *Drosophila* brain, which are instead slightly increased. We therefore searched for another neurotransmitter system that could be affected by *Appt* deficiency.

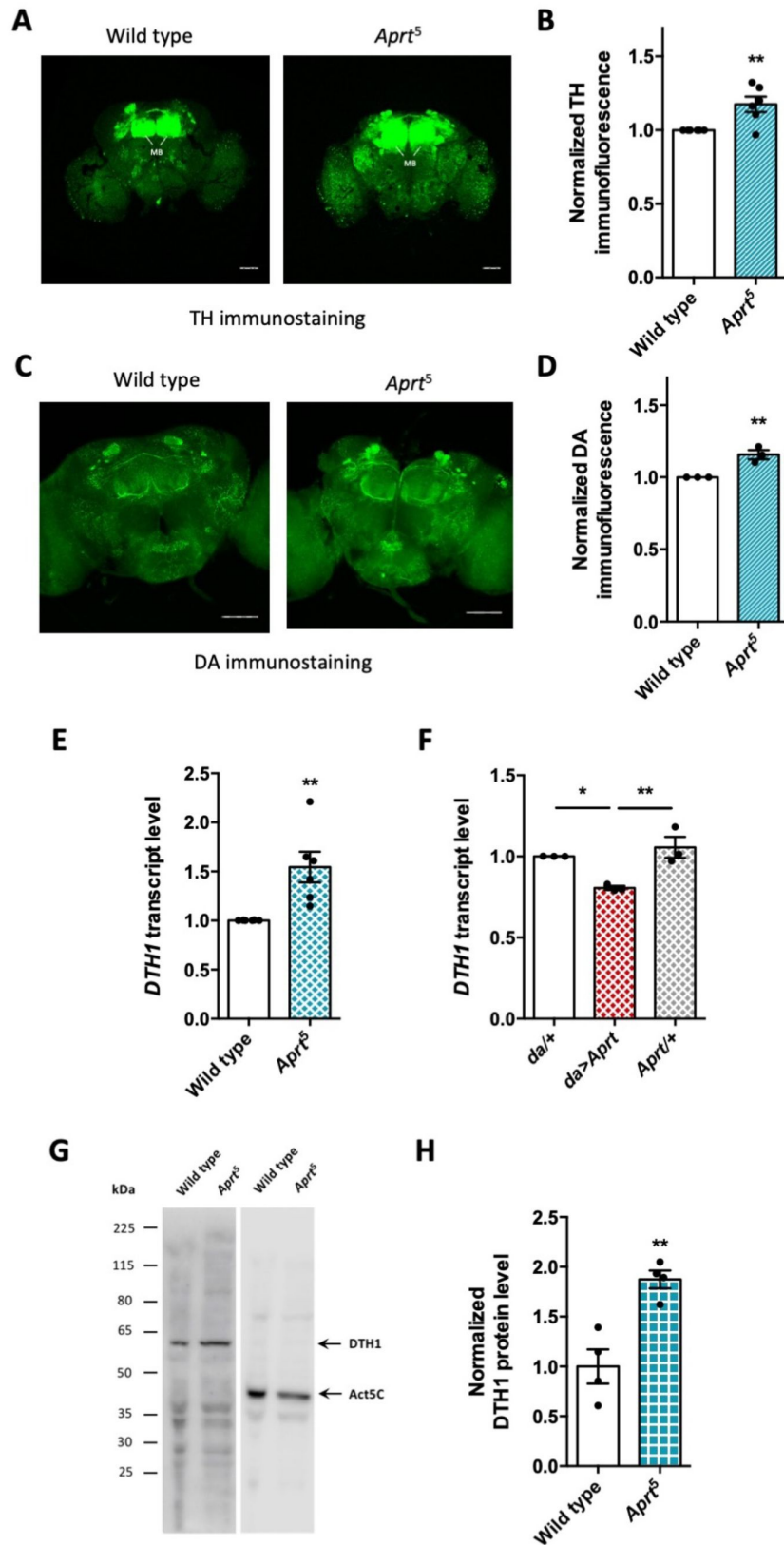


Figure 5.

***Aprt* deficiency increases DA synthesis and content in the *Drosophila* brain.**

(A) Representative confocal projections of TH-immunostained whole-mount adult brains from wild-type flies and *Aprt*⁵ mutants. MB: mushroom body. Scale bars: 100 μ m. (B) Quantification of TH immunofluorescence intensity normalized to the controls in the entire brain. 4 to 6 brains were dissected per experiment and genotype, and 6 independent experiments were performed (***p* < 0.01). (C) Representative confocal projections of DA immunostaining in whole-mount adult brains of wild type and *Aprt*⁵ mutants. Scale bars: 100 μ m. (D) Quantification of DA immunofluorescence intensity over the entire brain showed a slight increase in DA content in *Aprt*⁵ mutants compared to wild-type controls. 6 brains were dissected per experiment and genotype, and 3 independent experiments were performed. Unpaired Student's *t* test (***p* < 0.01). (E) mRNA levels of tyrosine hydroxylase neuronal form *DTH1* are markedly increased in *Aprt*⁵ mutant heads compared to wild-type flies. Results of 6 independent RT-qPCR experiments carried out on 3 to 4 different RNA extractions from 20-30 male heads per genotype. Unpaired Student's *t* test (***p* < 0.01). (F) Conversely, overexpressing *Aprt* in all cells with the *da-Gal4* driver (*da>Aprt*) reduced mRNA level of *DTH1* in heads, as compared to the driver (*da/+*) and effector (*Aprt/+*) controls. Mean of 3 independent experiments performed on 3 different RNA extractions from 20-30 male fly heads. One-way ANOVA with Tukey's *post-hoc* test for multiple comparisons (**p* < 0.05, ***p* < 0.01). (G) Representative western blot of wild-type and *Aprt*⁵ mutant head extracts probed with anti-TH and anti-actin beta antibodies. (H) Quantification of DTH1 protein levels in adult wild-type and *Aprt*⁵ mutant heads by western blots showed an increase in DTH1 protein level (60 kDa) in *Aprt*⁵ mutants. Actin (Act5C, 42kDa) was used as a loading control. Results are the mean of four determinations in two independent experiments. Unpaired Student's *t* test (***p* < 0.01).

Interactions between *Aprt* and the adenosinergic system

Aprt catalyzes the conversion of adenine into AMP, and AMP breakdown by the enzyme 5'-nucleotidase produces adenosine, primarily in the extracellular space. Adenosine then acts as a widespread neuromodulator in the nervous system by binding to adenosine receptors. We therefore suspected that adenosine level could be reduced in the absence of *Aprt* activity, leading to alterations in adenosinergic neurotransmission. Indeed, we found a significant decrease in adenosine level either in whole flies (by 61.0% on average, *p* < 0.01) or in heads (by 48.0%, *p* < 0.05) in *Aprt*⁵ mutants (Figure 6A). We then examined the consequences of this reduction on molecular components of the adenosinergic system, namely G protein-coupled adenosine receptors and equilibrative nucleoside transporters (ENTs), which carry out nucleobase and nucleoside uptake of into cells. Only one seven-transmembrane-domain adenosine receptor, *AdoR*, is present in *Drosophila*, that is very similar to mammalian adenosine receptors (Dolezelova et al. 2007; Brody and Cravchik, 2000), and three putative equilibrative nucleoside transporters (*Ent1-3*) have been identified (Sankar et al. 2002) but only one, *Ent2*, showed nucleoside transport activity (Machado et al. 2007). Interestingly, *Ent2* has been shown to be enriched in the mushroom bodies and its transcript level to be highly elevated in *AdoR* mutant flies (Knight et al. 2010). In *Aprt*⁵ background, we also observed a strong increase in *Ent2* mRNAs (2.3-fold higher than wild-type flies, *p* < 0.001), but no noticeable effect on *AdoR* transcript level (Figure 6B). The increased expression of *Ent2* could correspond to a compensatory response to adenosine shortage in *Aprt*⁵ mutants.

The *AdoR* receptor is highly expressed in adult fly heads and its ectopic overexpression leads to early larval lethality (Dolezelova et al. 2007). In contrast, a null mutant of this receptor, *AdoR*^{KGex}, in which the entire coding sequence is deleted, is fully viable (Wu et al. 2009). This enabled us to examine the consequences of a complete lack of *AdoR* on purine recycling in adult flies. We found that *Aprt* transcripts were decreased by 29.5% on average (*p* < 0.001) in *AdoR*^{KGex} mutant heads (Figure 6C, left panel), while *Aprt* activity was even more decreased by 78.4% (*p* < 0.001) compared to wild-type flies (Figure 6C, right panel). This effect likely results from the

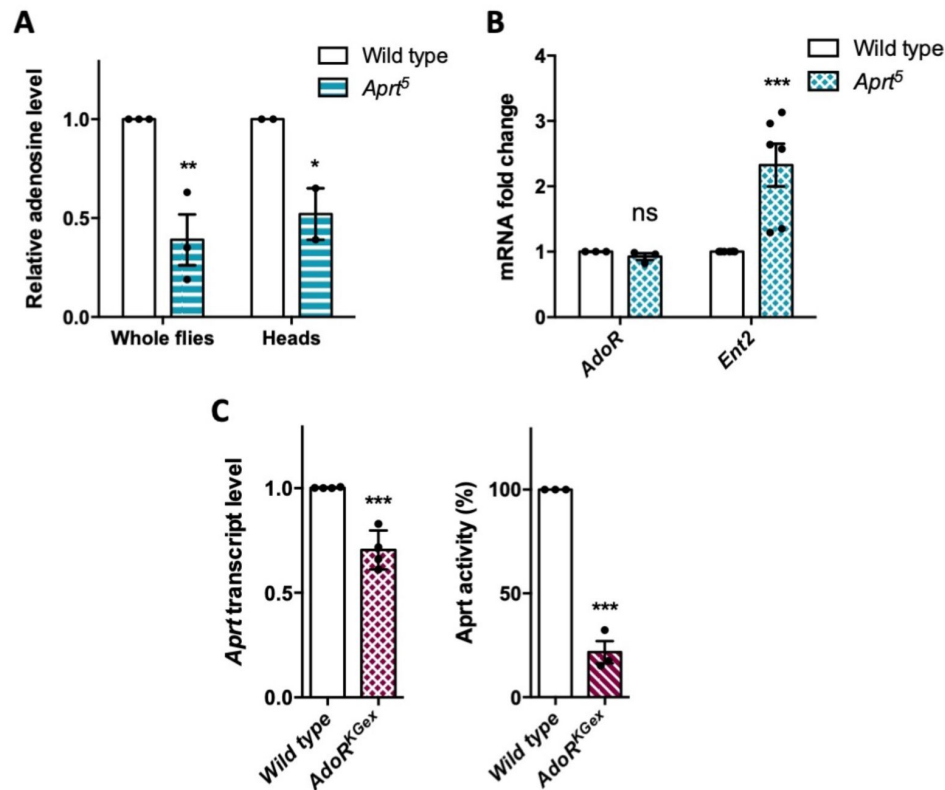


Figure 6.

Relations between *Aprt* and molecular components of adenosinergic signaling.

(A-B) Impacts of the lack of *Aprt* activity on the adenosinergic system.

(A) Adenosine level was measured in whole flies or heads of *Aprt⁵* flies by UPLC. Compared to wild-type flies, adenosine level was significantly reduced in the mutants. Results of 3 independent experiments performed with 5 males per genotype in triplicates, and 2 independent experiments with 30 heads per genotype in triplicates. Two-way ANOVA with Sidak's *post-hoc* test for multiple comparisons (* $p < 0.05$; ** $p < 0.01$). (B) *Aprt⁵* mutation did not affect *AdoR* expression but induced a marked increase in adenosine transporter *Ent2* mRNA abundance. 3-6 different RNA extractions were performed on 20-30 male heads. Results of 3 to 6 independent experiments. Two-way ANOVA with Sidak's *post-hoc* test for multiple comparisons (*** $p < 0.001$; ns: not significant). (C) Null *AdoR^{KGex}* mutants showed decreased *Aprt* expression (left panel) and a stronger decrease in *Aprt* activity (right panel). 4 independent RNA extractions were carried out on 20-30 male heads and 4 independent real-time PCR determinations were done per RNA sample. For *Aprt* activity, 3 independent determinations were performed on 20 whole flies per genotype. Unpaired Student's *t* test (*** $p < 0.001$).

much increased level of adenosine uptake in the *AdoR* mutants, which would downregulate *Aprt* activity by a feedback mechanism. Overall, these data indicate that extracellular adenosine levels must be strongly decreased in *Aprt* mutant flies, both from the general reduction in adenosine levels and the increased expression of the *Ent2* transporter. Although *AdoR* expression is not affected, *AdoR* signaling is therefore expected to be significantly reduced in *Aprt* mutants.

***Aprt* mutants show epilepsy-like seizure behavior**

A number of *Drosophila* mutants with disrupted metabolism or neural function show increased susceptibility to seizure and paralysis following strong mechanical or electrophysiological stimulation (Fergestad et al. 2006 [\[1\]](#), Parker et al. 2011 [\[2\]](#), Kroll et al. 2015 [\[3\]](#)). These mutants are called bang-sensitive (BS) and commonly used as models of epileptic seizure (Song and Tanouye 2008 [\[4\]](#)). Here we checked whether *Aprt* deficiency could trigger a BS phenotype. We observed that aged *Aprt* mutants (at 30 d a.E.) recovered slowly after a strong mechanical stimulation applied by vortexing the vial for 10 s at high speed. These flies took on average 17.3 s to recover and get back on their legs, compared to 2.5 s for wild-type flies of the same age ($p < 0.01$) (Figure 7A [\[5\]](#) and Movie S1). Some of the mutant flies appeared more deeply paralyzed as they did not spontaneously recover unless the vial was stirred, so their recovery time could not be scored. Hemizygous flies of the same age containing the *Aprt*⁵ mutation over two partially overlapping genomic deficiencies covering *Aprt* also showed a BS behavior, with an average longer recovery time of 28.9 s and 33.2 s, respectively ($p < 0.001$) (Figure 7B [\[6\]](#)).

We then downregulated *Aprt* by RNAi in all cells with the *da-Gal4* driver to check if this could also induce BS behavior. As shown in Figure 7C [\[7\]](#), *da>Aprt*^{RNAi} flies at 30 d a.E. indeed displayed seizure after mechanical shock, quite similar to that of the *Aprt*⁵ mutants (22.7 s recovery time on average compared to 1.55 s and 0.72 s for the driver and RNAi controls, respectively, $p < 0.05$). In contrast, inactivating *Aprt* selectively in neurons (*elav-Gal4*), glial cells (*repo-Gal4*) or muscles (*24B-Gal4*) did not induce a BS phenotype (Figure S13). This suggests that the BS phenotype requires *Aprt* knockdown in other cells or in several of these cell types. Finally, in contrast to the SING defect, we observed that adult-specific *Aprt* knockdown in all cells with *da-Gal4* for 3 days did not trigger a BS behavior (Figure 7D [\[8\]](#)), indicating that the bang sensitivity requires a longer downregulation of the gene or might be the consequence of a developmental defect.

Administration of Adenosine or *N*⁶-methyladenosine to *Aprt* $\mp$ deficient flies prevents seizure

Drosophila disease models are advantageously tractable for drug screening *in vivo* (Fernández-Hernández et al. 2016 [\[9\]](#), Perrimon et al. 2016 [\[10\]](#)). We thus administered several compounds related to purine metabolism to *Aprt*⁵ flies to check if they can rescue neurobehavioral impairments (locomotor defects and seizure). Feeding allopurinol at the same concentration used for uric acid normalization (100 μ g/ml, Figure 1B [\[11\]](#)), either in adults 5 days before the test or throughout all developmental stages, did not alter the BS phenotype (Figure S14), as was the case for the SING assay (Figure 1D [\[12\]](#), E). Similarly in humans, it has been shown that the daily intake of allopurinol, even from infancy, does not mitigate the neurobehavioral impairments in LND patients (Marks et al. 1968 [\[13\]](#); Torres et al. 2007 [\[14\]](#); Jinnah et al. 2010 [\[15\]](#); Madeo et al. 2019 [\[16\]](#)).

Then, we tried to supplement *Aprt* mutants with various purine compounds, including adenine, hypoxanthine, adenosine and *N*⁶-methyladenosine (*m*⁶A), at 100 or 500 μ M, either in adult stage 5 days before the test or throughout larval development plus 5 days before the test. None of these drugs was able to rescue the SING defect (Figure S15). In contrast and interestingly, administration of 500 μ M adenosine or *m*⁶A during development rescued the BS phenotype of *Aprt*⁵ mutants (Figure 7E-F [\[17\]](#)). This further indicates that different mechanisms underpin SING alteration and BS

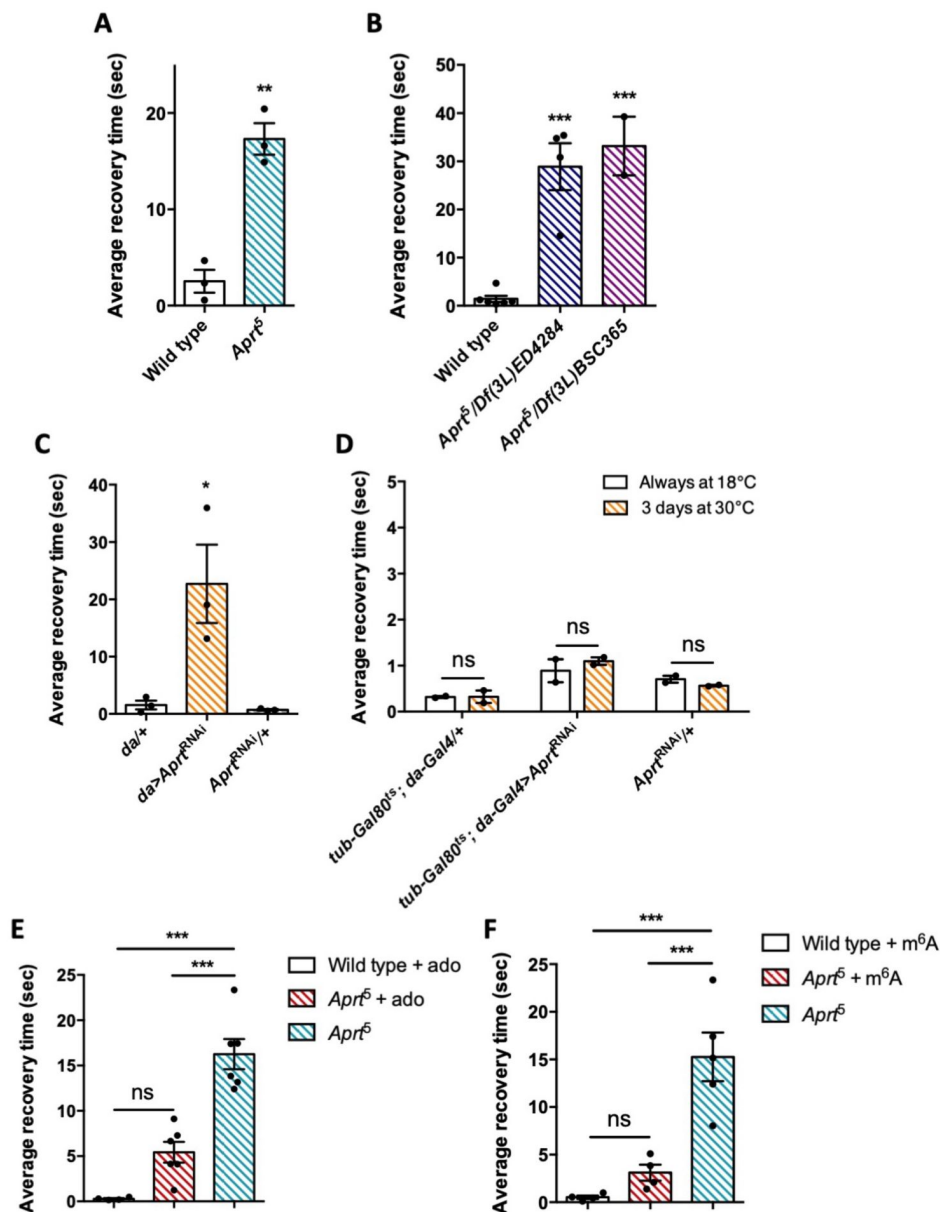


Figure 7.

***Aprt* deficiency triggers a seizure-like phenotype.**

(A) At 30 d a.E., *Aprt*⁵ mutants need a much longer time than wild-type flies to recover from a strong mechanical shock, showing a bang-sensitive (BS) paralysis comparable to tonic-clonic seizure. Results of 3 independent experiments performed on 50 flies per genotype. Unpaired Student's *t* test; ***p* < 0.01. (B) At 30 d a. E., hemizygous *Aprt*⁵ mutants also showed a marked BS phenotype. Results of 2-4 independent experiments performed on 50 flies per genotype. One-way ANOVA with Dunnett's *post-hoc* test for multiple comparisons (****p* < 0.001). (C) RNAi-mediated downregulation of *Aprt* in all cells (*da>Aprt*^{RNAi}) also led to BS phenotype in adults at 30 d a.E., but not with the driver and effector controls. Results of 3 independent experiments performed on 50 flies per genotype. One-way ANOVA with Tukey's *post-hoc* test for multiple comparisons (**p* < 0.05). (D) *Aprt* knockdown by RNAi during the adult stage for 3 days before the test was not sufficient to induce bang-sensitivity, suggesting that this phenotype could be caused by a developmental defect or a longer downregulation of *Aprt*. Results of 2 independent experiments performed on 50 flies per genotype. Two-way ANOVA with Sidak's *post-hoc* test for multiple comparisons; ns: not significant. (E, F) The BS phenotype of 30-day-old *Aprt*⁵ mutants was rescued by feeding either 500 μM adenosine (ado) (E) or 500 μM N⁶-methyladenosine (m⁶A) (F) during all developmental stages plus 5 days before the test. Results of 4 or 6 independent experiments performed on 50 flies per genotype. One-way ANOVA with Tukey's *post-hoc* test for multiple comparisons (****p* < 0.001, ns: not significant).

behavior in *Ap^{rt}* mutants and provide another evidence that the bang sensitivity may be caused by a developmental defect. The results also suggest that the lower adenosine levels of *Ap^{rt}* mutant flies could be at the origin of their bang sensitivity.

Neuronal expression of mutant HGPRT induces early locomotor defects and seizure behavior

In order to potentially develop another *Drosophila* model mimicking LND conditions, we generated new transgenic UAS lines to express in living flies either the human wild-type HGPRT (HGPRT-WT) or a pathogenic LND-associated mutant form of this protein (HGPRT I42T), both isoforms being inserted at the same genomic docking site. These lines were validated by showing that they are transcribed at similar levels (**Figure 8A** [↗](#), B). Enzymatic assays on adult extracts of flies expressing the wild-type form HGPRT-WT in all cells showed significant HGPRT enzyme activity, while an 80.5% lower activity was detected in *Drosophila* expressing the mutant form HGPRT-I42T (Table S2).

We next analyzed the consequences of human HGPRT expression on the SING and bang sensitivity behaviors. Interestingly, the pan-neuronal expression of mutant I42T isoform specifically induced a significant early locomotor defect at 15 d a.E. (PI = 0.71 vs. 0.92 and 0.90 for the driver and effector controls, respectively, $p < 0.01$) (**Figure 8C** [↗](#)) and a relatively small but robust bang-sensitivity behavior at 30 d a.E. (2.3 s average recovery time vs. 0.64 s and 0.31 s for the driver and effector only controls, $p < 0.001$) (**Figure 8D** [↗](#)). These defective phenotypes could not be seen when HGPRT-WT was expressed. Therefore, and remarkably, whereas wild-type HGPRT expression appears to be innocuous in *Drosophila*, we observed that the neuronal expression of a pathogenic LND-associated isoform triggered neurobehavioral impairments quite similar to those of *Ap^{rt}*-deficient flies.

Discussion

Over the past 35 years, several animal models of LND have been developed in rodents based on HGPRT mutation in order to better understand the cause of the disease and test potential therapeutic treatments (Finger et al. 1988 [↗](#); Dunnett et al. 1989 [↗](#); Jinnah et al. 1993 [↗](#); Engle et al. 1996 [↗](#); Meek et al. 2016 [↗](#); Witteveen et al. 2022 [↗](#)). However, none of these models recapitulated the full LND syndrome and, particularly, motor or neurobehavioral symptoms as a consequence of HGPRT deficiency. To date, the causes of the neurobehavioral impairments in LND are not yet clearly elucidated and the disease is still incurable (Fu et al. 2014 [↗](#); Bell et al. 2016 [↗](#); López et al. 2020 [↗](#); Bell et al. 2021 [↗](#); Witteveen et al. 2022 [↗](#)). Here we used two different strategies to develop new models of this disease in *Drosophila*, a useful organism to conduct genetic and pharmacological studies. First, we show that *Ap^{rt}* deficiency induces both metabolic and neurobehavioral disturbances in *Drosophila*, similar to the loss of HGPRT, but not APRT, activity in humans. Secondly, we expressed an LND-associated mutant form of human HGPRT in *Drosophila* neurons, which also yielded behavioral symptoms. Our results suggest that the fruit fly can be used to study the consequences of defective purine recycling pathway and HGPRT mutation in the nervous system.

Ap^{rt}-deficient flies replicate lifespan and metabolic defects caused by HGPRT deficiency

Flies that carry a homozygous null-mutation in *Ap^{rt}* develop normally and live until the adult stage (Johnson and Friedman, 1983 [↗](#)). While a previous study reported that heterozygous *Ap^{rt}/+* flies have an extended lifespan (Stenesen et al. 2013 [↗](#)), we observed that homozygous *Ap^{rt}⁵* mutants have in contrast a significantly reduced longevity. The lack of HGPRT activity in LND also reduces lifespan expectancy, generally under 40 years of age for properly cared patients. Stenesen

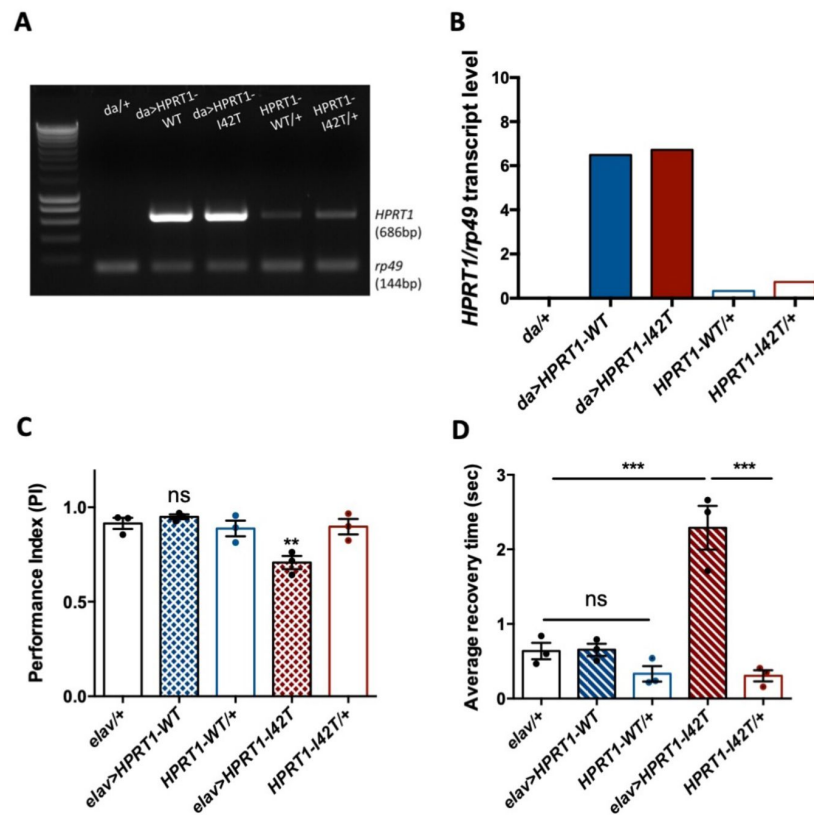


Figure 8.

Expression of a pathogenic mutant isoform of human HGPRT induces neurobehavioral defects in flies.

(A-B) Ubiquitous expression of human *HPRT1* with *da-Gal4*. (A) Amplification of human *HPRT1* transcripts detected by RT-PCR in head extracts of *da>HPRT1-WT* and *da>HPRT1-I42T* flies. A band with lower intensity was also detected in the effector controls (*UAS-HPRT1-WT/+* and *UAS-HPRT1-I42T/+*), and not in the driver control (*da/+*), which indicates a small amount of driver-independent transgene expression. (B) Quantification of the previous experiment. (C, D) Expression of the LND-associated I42T isoform in all neurons (*elav>HPRT1-I42T*), but not of the wild-type form (*elav>HPRT1-WT*), induced an early locomotor SING defect at 15 d a.e. (C) and a prominent bang-sensitivity phenotype at 30 d a.e. (D), as compared to the driver (*elav/+*) and effector (*UAS-HPRT1 I42T/+*) only controls. Results of 3 independent experiments performed on 50 flies per genotype. One-way ANOVA with Tukey's *post-hoc* test for multiple comparisons (** $p < 0.01$; *** $p < 0.001$; ns: not significant).

et al. (2013) [\[13\]](#) showed that dietary supplementation with adenine, the *Appt* substrate, prevented the longevity extension conferred either by dietary restriction or heterozygous mutations of AMP biosynthetic enzymes. This suggests that lifespan depends on accurate adenine level regulation. It is possible that adenine could accumulate to toxic levels during aging in homozygous *Appt* mutants, explaining their shorter lifespan. Alternatively, since AMP is the *Appt* product, AMP-activated protein kinase (AMPK), an enzyme that protects cells from stresses inducing ATP depletion, could be less activated in *Appt* mutants. Multiple publications explored the role of AMPK in lifespan regulation (Sinnott and Brenman 2016 [\[14\]](#)) and downregulating AMPK by RNAi in adult fat body or muscles (Stenesen et al. 2013 [\[15\]](#)), as well as its ubiquitous inactivation under starvation (Johnson et al. 2010 [\[16\]](#)), both reduce fly lifespan.

In humans, HGPRT deficiency induces hypoxanthine and guanine accumulation, resulting from lack of recycling, and increased *de novo* purine synthesis (Harkness et al. 1988 [\[17\]](#), Fu et al. 2015 [\[18\]](#); Ceballos-Picot et al. 2015 [\[19\]](#)). This in turn leads to uric acid overproduction that increases the risk for nephrolithiasis, renal failure and gouty arthritis if not properly treated. In insects, the end-product of purine catabolism is not uric acid but allantoin (Figure S1, step 20). However, urate oxidase, the enzyme which converts uric acid into allantoin, is specifically expressed in the Malpighi tubules, an excretory organ producing pre-urine (Wallrath et al. 1990 [\[20\]](#)). Uric acid could therefore accumulate in fly tissues and hemolymph if purine salvage pathway is impaired. Accordingly, we observed an increase in uric acid levels in *Drosophila Appt* mutant heads, which could be reduced to normal levels by providing allopurinol, a xanthine oxidase inhibitor used to protect against renal failure in LND patients.

Appt is required in dopamine and mushroom body neurons for young fly motricity

We found that *Appt*-null adult flies show reduced performance in the SING test, which monitors locomotor reactivity to startle and climbing ability. This phenotype appears at an early age, starting from 1 d a.E. The performance continued to decrease until 8 d a.E. and then appeared to stabilize up to 30 d a.E. This defect is quite different from the locomotor impairments described in *Drosophila* Parkinson's disease models, in which SING performance starts declining at around 25 d a.E. (Feany and Bender 2000 [\[21\]](#), Riemensperger et al. 2013 [\[22\]](#), Rahmani et al. 2022 [\[23\]](#)). This phenotype of *Appt*-deficient flies could be reminiscent of the early onset of motor symptoms in LND patients, which appear most often between 3 to 6 months of age (Jinnah et al. 2006). Interestingly, knocking down *Appt* during 3 days only in adult flies also induced SING impairment, which argues against a developmental flaw. Although *Appt* mutants walked slower than wild-type flies, they were no less active and their ATP levels were not different compared to controls, excluding a major failure in energy metabolism.

Downregulating *Appt* in all neurons reproduced the locomotor defect of the *Appt*⁵ mutant, and *Appt* mutant locomotion could be partially rescued by neuronal *Appt* re-expression. The fact that rescue was not complete could be due to a dominant negative effect of the mutation, as suggested by enzymatic assays on extracts of heterozygous *Appt*⁵ mutants (Figure S4). Previous work from our and other laboratories showed that DA neurons control the SING behavior in *Drosophila* (Feany and Bender 2000 [\[21\]](#); Friggi-Grelin et al. 2003 [\[24\]](#); Riemensperger et al. 2011 [\[25\]](#); Vaccaro et al. 2017 [\[26\]](#); Sun et al. 2018 [\[27\]](#)) and that PAM DA neuron degeneration induce SING defects in various Parkinson's disease models (Riemensperger et al. 2013 [\[22\]](#); Bou Dib et al. 2014 [\[28\]](#); Tas et al. 2018 [\[29\]](#); Pütz et al. 2021 [\[30\]](#)). Here, we found that knocking down *Appt* either in all DA neurons or only in the PAM DA neurons was sufficient to induce early SING defects. In contrast, knocking down *Appt* with *TH-Gal4* that labels all DA neurons except for a major part of the PAM cluster did not induce SING defects, indicating that *Appt* expression in subsets of PAM neurons is critical for this locomotor behavior.

Inactivating *Appt* in all MB neurons also induced a lower performance in the SING assay. This important brain structure receives connections from DA neurons, including the PAM, and is enriched in DA receptors (Waddell 2010 [↗](#)). We recently reported that activation of MB-afferent DA neurons decreased the SING response, an effect that requires signalization by the DA receptor dDA1/Dop1R1 in MB neurons (Sun et al. 2018 [↗](#)). We also observed a SING defect after knocking down *Appt* either in all glial cells, or more selectively in glial subpopulations expressing the glutamate transporter *Eaat1*, but not in the ensheathing glia. Astrocyte-like glial cells expressing *Eaat1* extend processes forming a thick mesh-like network around and inside the entire MB neuropil (Sinakevitch et al. 2010 [↗](#)). It could be hypothesized that SING also requires *Appt* expression in these MB-associated astrocytes. However, re-expressing *Appt* with *Eaat1-Gal4* did not lead to SING rescue in *Appt* mutant background. This suggests that the presence of *Appt* in neurons can somehow compensate for *Appt* deficiency in glia, but the reverse is not true. In conclusion, proper startle-induced locomotion in young flies depends on *Appt* activity in PAM DA and MB neurons, and in *Eaat1*-expressing glial cells.

Neuronal *Appt* regulates spontaneous activity and sleep

Because sleep is regulated by the mushroom body as well as DA neurons in flies, (Artiushin and Sehgal 2017 [↗](#)), we monitored spontaneous locomotor activity and sleep pattern of *Appt*⁵ mutants. Their activity profile appeared normal, with unaltered morning and evening anticipation, indicating that the circadian rhythms are surprisingly maintained in light-dark conditions in the absence of a functional purine recycling pathway (Figure S9). This experiment also highlighted that *Appt*-deficient flies are hyperactive during the night, which suggests that they sleep less than wild-type flies. This could be confirmed by measuring their sleep pattern on video recordings. *Appt* mutants show a reduced walking speed, sleep less during both day and night, and have a difficulty to maintain sleep. Downregulating *Appt* selectively in neurons reproduced the sleep defect, whereas doing it in glial cells only had no effect. We have not attempted here to identify further the neuronal cells involved in this phenotype. Like for the SING behavior defect, it could involve PAM DA neurons, as subpopulations of this cluster have been shown to regulate sleep in *Drosophila* (Nall et al. 2016 [↗](#)). Therefore, the lack of purine recycling markedly disrupts sleep in *Appt*-deficient flies, making them more active at night. This is strikingly comparable to young LND patients who have a much disturbed sleep-time during the night (Mizuno et al. 1979 [↗](#)).

Lack of *Appt* reduces adenosine signaling leading to DA neuron overactivation

We observed that *Appt* deficiency did not decrease DA levels in the *Drosophila* brain. This prompted us to study another neuromodulator, adenosine, which is indirectly a product of *Appt* enzymatic activity. The purine nucleoside adenosine is one of the building blocks of RNA and the precursor of ATP and cAMP, but is also the endogenous ligand of adenosine receptors that modulate a wide range of physiological functions. In brain, adenosine regulates motor and cognitive processes, such as the sleep-wake cycle, anxiety, depression, epilepsy, memory and drug addiction (Soliman et al. 2018 [↗](#)). In normal metabolic conditions, adenosine is present at low concentrations in the extracellular space and its level is highly regulated, either taken up by cells and incorporated into ATP stores or deaminated by adenosine deaminase into inosine. In mammals, several nucleoside transporters mediate the uptake of adenosine and other nucleosides into cells, named equilibrative and concentrative nucleoside transporters (ENTs and CNTs), respectively (Gray et al. 2004 [↗](#); Boswell-Casteel and Hays 2017 [↗](#); Pastor-Anglada and Pérez-Torrás 2018 [↗](#)).

We observed a marked reduction in adenosine levels in *Appt* mutant flies. While we did not observe any alteration in the transcript levels of *AdoR*, the gene coding for the only adenosine receptor in *Drosophila*, transcript levels of an adenosine transporter, *Ent2*, were increased more than 2-fold in *Appt* mutants. Interestingly, one paper reported a similar strong increase in *Ent2* expression in *AdoR* mutant flies (Knight et al. 2010 [↗](#)). These results suggest that *AdoR* signaling is

less activated in *Appt*-deficient flies compared to controls. We also observed that the lack of AdoR decreased *Appt* expression and activity in *Drosophila*, possibly from increased *Ent2* expression and so higher adenosine influx which could downregulate *Appt* by a feedback mechanism.

Adenosine and dopamine receptors are known to interact closely in mammals (Franco et al. 2000; Kim and Palmiter 2008). A previous study performed in fly larvae showed that an increase in astrocytic Ca^{2+} signaling can silence DA neurons through AdoR stimulation, by a mechanism potentially involving the breakdown of released astrocytic ATP into adenosine (Ma et al. 2016). We previously showed that DA neuron activation can decrease fly performance in the SING test (Sun et al. 2018). Fly nocturnal hyperactivity can also be caused by an increase in dopamine signaling (Kumar et al. 2012; Lee et al. 2013), in accordance with our observation that *Appt*-deficient flies sleep less and are more active during the night. Therefore, both the locomotor and sleep defects induced by the lack of *Appt* activity could be explained by DA neuron overactivation that would result from reduced adenosinergic signaling.

Adenosine has been proposed before to be involved in neurological consequences of LND (Visser et al. 2000). Adenosine transport is decreased in peripheral blood lymphocytes of LND patients (Torres et al. 2004), as well as $\text{A}_{2\text{A}}$ adenosine receptor mRNA and protein levels (García et al. 2009; García et al. 2011). In a murine LND model, adenosine A_1 receptor expression was found to be strongly increased and that of $\text{A}_{2\text{A}}$ slightly decreased, while $\text{A}_{2\text{B}}$ expression was not affected (Bertelli et al. 2006). Chronic administration of high doses of caffeine, an adenosine receptor antagonist, caused self-injurious behavior in rats (Miñana et al. 1984; Jinnah et al. 1990). Moreover, central injection of an adenosine agonist is sufficient to prevent self-mutilation induced by dopaminergic agonist administration in neonatally 6-OHDA lesioned rats (Criswell et al. 1988).

Adenosine or N6-methyladenosine supplementation rescues the epilepsy behavior of *Appt* mutants

LND is characterized by severe behavioral troubles including dystonia, spasticity and involuntary movements (choreoathetosis). Some patients can also show an epileptic disorder (Jinnah et al. 2006; Madeo et al. 2019). In flies, the bang-sensitivity (BS) test is often used to model epileptic seizures (Song et al. 2008; Parker et al. 2011). Here, we observed that 30-day-old *Appt* mutant flies show a transient seizure-like phenotype after a strong mechanical shock. Although seizure duration appeared shorter in *Appt*⁵ than in typical BS mutants such as *bss*, at least one other BS mutant, *porin*, was reported to have similar short recovery times as *Appt*-deficient flies (Graham et al. 2010). Previous works demonstrated that bang sensitivity is linked to neuronal dysfunction in *Drosophila* (Parker et al. 2011; Kroll and Tanouye 2013; Kroll et al. 2015; Saras and Tanouye 2016). Knocking down *Appt* in specific cells such as neurons, glia, or muscles, did not trigger this phenotype, suggesting that *Appt* must be inactivated in several cell types to induce seizure.

Interestingly, knocking down *Appt* by RNAi in all cells during development also induced the BS behavior, but not for 3 days only in adult flies, at variance with the SING phenotype. We have fed the mutants with a diet supplemented with various compounds involved in purine metabolism, including allopurinol, adenine, hypoxanthine, adenosine or its analog N6-methyladenosine (m^6A), either throughout larval development or in adult stage. Only adenosine and m^6A , ingested during development, rescued the BS behavior. This suggests that loss of *Appt* induces a lack of adenosine in the developing nervous system, as we observed in adult flies (Figure 6), which may alter neural circuits controlling BS behavior in adults. The adenosine analogue m^6A cannot be incorporated into nucleic acids and is excreted in the urine (Schram 1998; Batista 2017). The rescuing effect we observed with m^6A suggests thereby that both this compound and adenosine are required as adenosine receptor agonists or allosteric regulators during development, rather than nucleotide precursors.

Adenosine can strongly inhibit cerebral activity and its role as endogenous anticonvulsant and seizure terminator is well-established in humans (Boison 2005 [↗](#); Masino et al. 2014 [↗](#); Weltha et al. 2019 [↗](#)). Conversely, deficiencies in the adenosine-based neuromodulatory system can contribute to epileptogenesis. For instance, increased expression of astroglial adenosine kinase (ADK), which converts adenosine into AMP, leads to a reduction in brain adenosine level that plays a major role in epileptogenesis (Weltha et al. 2019 [↗](#)). Hence, therapeutic adenosine increase is a rational approach for seizure control. Our observation that feeding adenosine or its derivative m^6A rescued the seizure-like phenotype of *Aprt* mutant flies further suggests that adenosinergic signaling has partly similar functions in the fly and mammalian brains. In addition, the decrease in adenosine levels we observed in *Aprt* mutants could result from enhanced ADK activity that would compensate for the lack of *Aprt*-produced AMP.

We and others recently observed that m^6A , and related compounds sharing an adenosine moiety, are able to rescue the viability of LND fibroblasts and neural stem cells derived from induced pluripotent stem cells (iPSCs) of LND patients cultured in the presence of azaserine, a potent blocker of *de novo* purine synthesis (Petitgas and Ceballos-Picot, unpublished results; Ruillier et al. 2020 [↗](#)). Like in flies again, allopurinol was not capable of rescuing the cell viability in this *in vitro* model. The similarity of these results does increase confidence that *Aprt*-deficient *Drosophila* could be used as an animal model of LND.

Expression of mutant HGPRT triggers locomotor and seizure phenotypes as does *Aprt* deficiency

How HGPRT deficiency can cause such dramatic neurobehavioral troubles in LND patients still remains a crucial question. To date, cellular (Smith & Friedman 2000; Torres et al. 2004 [↗](#); Ceballos-Picot et al. 2009 [↗](#); Cristini et al. 2010 [↗](#); Guibinga et al. 2012 [↗](#); Sutcliffe et al. 2021 [↗](#)) and rodent (Finger et al. 1988 [↗](#); Dunnett et al. 1989 [↗](#); Jinnah et al. 1993 [↗](#); Meek et al. 2016 [↗](#); Witteveen et al. 2022 [↗](#)) models only focused on the consequences of HGPRT deficiency to phenocopy the disease. Such an approach was justified by the fact that the lower the residual activity of mutant HGPRT, the more severe the symptoms are in patients (Fu and Jinnah 2012 [↗](#); Fu et al. 2014 [↗](#)). However, it could be conceivable that part of these symptoms result from compensatory physiological mechanisms or a deleterious gain-of-function conferred to the HGPRT protein by the pathogenic mutations. Here we observed that neuronal expression of mutant human HGPRT-I42T, which expresses a low enzymatic activity, but not the wild-type HGPRT protein, induced early locomotor defects in young flies and bang-sensitivity in older flies, similarly to the defects induced by *Aprt* deficiency. This suggests a potential neurotoxicity of the pathological mutant form of HGPRT, which could be related to disturbances in purine metabolism or other signaling pathways. The human mutant form might not be properly degraded and accumulate as aggregates, potentially exerting neuronal toxicity. Such an approach opens interesting perspectives to better understand the endogenous function of HGPRT and its pathogenic forms. Indeed, the identification of a potential inherent neurotoxicity of defective forms of human HGPRT is a new element, which could be explored in further work in the fly and also in rodent models.

A new model of LND in an invertebrate organism?

LND, a rare X-linked metabolic disorder due to mutations of the *HPRT1* gene, has dramatic neurological consequences for affected children. To date, no treatment is available to abrogate these troubles, and no fully satisfactory *in vivo* models exist to progress in the understanding and cure of this disease. HGPRT knockout rodents do not show comparable motor and behavioral troubles, which makes these models problematic for testing new therapeutic treatments. *Drosophila* does not express HGPRT-like activity and our phylogenetic analysis established that no HGPRT homologue is present in *Drosophila melanogaster* (Figure S2 and S3), confirming that *Aprt* is the only enzyme of the purine salvage pathway in this organism. APRT and HGPRT are known to be functionally and structurally related. Both human APRT and HGPRT belong to the type I

PRTases family identified by a conserved phosphoribosyl pyrophosphate (PRPP) binding motif, which is used as a substrate to transfer phosphoribosyl to purines. This binding motif is only found in PRTases from the nucleotide synthesis and salvage pathways (Sinha and Smith 2001 [\[1\]](#)). Moreover, the purine substrates adenine, hypoxanthine and guanine share the same chemical skeleton and APRT can bind hypoxanthine, indicating that APRT and HGPRT also share similarities in their substrate binding sites (Ozeir et al. 2019 [\[2\]](#)).

Here we find that *Aprt* mutant flies show symptoms partly comparable to the lack of HGPRT in humans, including increase in uric acid levels, reduced longevity, and various neurobehavioral defects such as early locomotor decline, sleep disorders and epilepsy behavior. This animal model therefore recapitulates both salvage pathway disruption and motor symptoms, as observed in LND patients. Moreover, these results highlighted that *Aprt* deficiency in *Drosophila* has more similarities with HGPRT than APRT deficiency in humans. *Aprt* mutant flies also show a disruption of adenosine signaling and we found that adenosine itself or a derivative compound can relieve the epileptic symptoms. Finally, neuronal expression of a mutant form of human HGPRT that causes LND also triggers abnormalities in fly locomotion and seizure-like behavior, which has not been documented to date in other models. The use of *Drosophila* to study LND could therefore prove valuable to better understand the link between purine recycling deficiency and brain functioning and carry out drug screening in a living organism, paving the way towards new improvements in the cure of this dramatic disease.

Materials and Methods

Drosophila culture and strains

Flies were raised at 25°C on standard cornmeal-yeast-agar medium supplemented with methylhydroxybenzoate as an antifungal under a 12h:12h light-dark (LD) cycle. The *Drosophila* mutant lines were obtained either from the Bloomington *Drosophila* Stock Center (BDSC), the Vienna *Drosophila* Resource Center (VDRC) or our own collection: *Aprt*⁵ (Johnson and Friedman, 1983 [\[3\]](#)) (BDSC #6882), *Df(3L)ED4284* (BDSC #8056), *Df(3L)BSC365* (BDSC #24389), *UAS-Aprt*^{RNAi} (VDRC #106836), *AdoR*^{KG03964ex} (Wu et al. 2009 [\[4\]](#)) here named *AdoR*^{KGex} (BDSC #30868), and the following Gal4 drivers: *238Y-Gal4*, *24B-Gal4*, *da-Gal4*, *Eaat1-Gal4* (Rival et al. 2004 [\[5\]](#)), *elav-Gal4* (*elav*^{C155}), *MZ0709-Gal4*, *NP6510-Gal4*, *NP6520-Gal4*, *R58E02-Gal4* (Liu et al. 2012 [\[6\]](#)), *R76F05-Gal4*, *repo-Gal4*, *TH-Gal4* (Friggi-Grelín et al. 2003 [\[7\]](#)), *TRH-Gal4* (Cassar et al. 2015 [\[8\]](#)), *tub-Gal80^{ts}* and *VT30559-Gal4*. The Canton-S line was used as wild-type control. The simplified driver>effector convention was used to indicate genotypes in figures, e.g. *elav>Aprt*^{RNAi} for *elav-Gal4*; *UAS-Aprt*^{RNAi}. In some experiments, to restrict RNAi-mediated *Aprt* inactivation at the adult stage, we have used the TARGET system (McGuire et al. 2003 [\[9\]](#)). Flies were raised at 18°C (permissive temperature) where Gal4 transcriptional activity was inhibited by *tub-Gal80^{ts}*, and shifted to 30°C (restrictive temperature) for 3 days before the test to enable Gal4-driven *Aprt*^{RNAi} expression.

Construction of transgenic lines

To generate a *UAS-Aprt* strain, a 549-bp *Aprt* insert containing the coding sequence was PCR amplified from the ORF clone BS15201 (*Drosophila* Genomics Resource Center, Bloomington, IN, USA) using primers with added restriction sites (Table S3). After digestion with EcoRI and XbaI, the *Aprt* cDNA was subcloned into the *pUASTattB* vector (Bischof et al. 2007 [\[10\]](#)) using the In-Fusion HD Cloning Kit (Takara Bio, Kyoto, Japan) according to the manufacturer's instructions, and the insertion verified by sequencing (Eurofins Genomics, Ebersberg, Germany). The construction was sent to BestGene (Chino Hills, CA, USA) for *Drosophila* germline transformation into the attP14 docking site on the 2d chromosome. The *UAS-Aprt* line yielding the strongest expression was selected and used in the experiments.

A clone containing the human wild-type *HPRT1* cDNA was kindly provided to us by Prof. Hyder A. Jinnah (Emory University, GA, USA). We constructed the *HPRT1-I42T* cDNA from this clone by site-directed mutagenesis using the Quick Change II Site-Directed Mutagenesis Kit (Agilent Technologies, Santa Clara, CA, USA). Primers sequences used to create the mutation are shown in Table S3. The cDNA obtained was verified by sequencing. Then, the 657-bp *HPRT1-WT* and *HPRT1-I42T* inserts were PCR amplified using primers with added restriction sites and *Drosophila* translation start consensus sequence (Table S3). They were subcloned into *pUASTattB* and verified by sequencing. The transgenes were sent to BestGene for *Drosophila* transformation and inserted into the attP40 docking site on the 2d chromosome.

Sequencing of *Aprt*⁵

For sequencing of the *Aprt*⁵ cDNA, total RNA was isolated by standard procedure from 20–30 heads of homozygous *Aprt*⁵ flies and reverse transcribed using oligo d(T) primers (PrimeScript RT Reagent Kit, Takara Bio, Kyoto, Japan). At least 750 ng of the first strand cDNA was amplified by PCR in 20 µl of reaction mixture using PrimeStar Max DNA polymerase (Takara Bio, Kyoto Japan) with a Techne Prime Thermal Cycler apparatus (Bibby Scientific, Burlington, NJ, USA). The program cycles included 10 s denaturation at 98°C, 10 s annealing at 55°C and 10 s elongation at 72°C, repeated 35 times. 1 µl of the PCR product was amplified again with the same program, in 30 µl of reaction mixture. After elution on 1% agarose gel, DNA were purified using the Wizard® SV Gel and PCR Clean-Up System protocol (Promega, Madison, Wisconsin, USA) according to the manufacturer's instructions. Finally, 7.5 µl of purified DNA were sent with 2.5 µl of primers (sense and antisense in separate tubes) for sequencing (Eurofins Genomics, Ebersberg, Germany).

Phylogenetic analyses

HGPRT homologs were identified by BlastP searching the NCBI GenBank non-redundant protein database (last October 2019 version). A subset of interest was selected for phylogenetic analyses. For Figure S2, multiple sequence alignment was performed using MAFFT (-ensi) (Kato and Standley 2013). The confidence of aligned residues was assessed using the TCS index (Chang et al. 2014); only columns with TCS index 7–9 (on a 0–9 scale) were retained. ProtTest v3.2 (Darriba et al. 2011) was used to assess the best model fitting of the data. Maximum Likelihood tree was inferred in IQ-TREE (Trifinopoulos et al. 2016), under the LG + R6 model. Bayesian inference was carried out in PhyloBayes v. 3.3 (Lartillot et al. 2009), under the LG + ρ 4 model. For Figure S3, the whole analysis was performed in SeaView 5.0.4 (Gouy et al. 2010). Multiple sequence alignment was performed using MUSCLE (Edgar 2004). Maximum likelihood tree was inferred using PhyML 3.0 (Guindon et al. 2010) (under the LG + ρ 4 model; best of NNI & SPR tree searching option; invariable sites optimized).

Lifespan assay

Longevity study was performed as previously described (Riemensperger et al. 2011). *Drosophila* adult males were collected within 24 h of emergence and maintained on standard medium at 25°C under a 12:12 h LD cycle. They were transferred into fresh bottles every 2–3 days and the number of surviving flies was scored. 50 flies per bottle in triplicate were tested for each genotype and the experiment was done 3 times.

Startle-induced negative geotaxis (SING)

SING assays were performed as previously described (Rival et al. 2004; Riemensperger et al. 2011; Sun et al. 2018). For each genotype, 50 adult males divided into 5 groups of 10 flies were placed in a vertical column (25-cm long, 1.5-cm diameter) with a conic bottom end and left for about 30 min for habituation. Then, columns were tested individually by gently tapping them down (startle), to which flies normally responded by climbing up. After 1 min, flies having reached at least once the top of the column (above 22 cm) and flies that never left the bottom (below 4 cm)

were counted. Each fly group was assayed three times at 15 min intervals. The performance index (PI) for each column is defined as $\frac{1}{2}[1 + (n_{\text{top}} - n_{\text{bot}})/n_{\text{tot}}]$, where n_{tot} is the total number of flies in the column, n_{top} the number of flies at the top and n_{bot} the number of flies at the bottom after 1 min. SING was performed at 1, 8, 10 and 30 days after adult eclosion (d a.E.).

Spontaneous locomotion and sleep monitoring

Spontaneous locomotor activity was recorded as previously described (Vaccaro et al. 2017 [↗](#)), using *Drosophila* activity infra-red beam monitors (DAM, TriKinetics Inc, Waltham, MA USA) placed in incubators at 25°C equipped with standard white light. 8-day-old male flies were maintained individually for 5-6 days under 12:12h LD cycle in 5 x 65 mm glass tubes containing 5% sucrose, 1.5% agar medium. Data analysis was performed with the FaasX software (Klarsfeld et al. 2003 [↗](#)). Histograms represent the distribution of the activity through 24h in 30-min bins, averaged for 32 flies per genotype over 4-5 cycles.

For sleep monitoring, 2-4-day-old virgin female flies were transferred individually into 5 x 65 mm glass tubes containing standard food and their movements were recorded for up to 5 days using DAM infra-red beam monitors (TriKinetics Inc, Waltham, MA USA) or the *Drosophila* Arousal Tracking (DART) video system (Faville et al. 2015 [↗](#)), in a 12h:12h LD cycle with 50-60% humidity. Control and experimental flies were recorded simultaneously. Each experiment included at least 14 flies for each condition and was repeated 2-3 times with independent groups of flies. Fly sleep, defined as periods of immobility lasting more than 5 min (Huber et al. 2004 [↗](#)), was computed with a Microsoft Excel macro for the infra-red beam data and with the DART software for the video data. Distribution and homogeneity as well as statistical group comparisons were tested using Microsoft Excel plugin software StatEL. The *p* value shown is the highest obtained among *post-hoc* comparisons and means $\pm$ SEM were plotted.

Immunohistochemistry

Brains of 8 to 10-day-old adult flies were dissected in ice-cold Ca^{2+} -free *Drosophila* Ringer's solution and processed for whole-mount anti-TH or anti-DA immunostaining as previously described (Riemensperger et al. 2011 [↗](#), Cichewicz et al. 2017 [↗](#)). The primary antibodies used were mouse monoclonal anti-TH (1:1000, cat. n° 22941, ImmunoStar, Hudson, WI, USA) and mouse monoclonal anti-DA (1:100, cat. n° AM001, GemacBio, Saint-Jean-d'Ilac, France). The secondary antibody was goat anti-mouse conjugated to Alexa Fluor 488 (1:1000, ThermoFisher Scientific, Waltham, MA, USA). Brains were mounted with antifade reagent, either Prolong Gold (ThermoFisher Scientific, Waltham, MA, USA) or, alternatively, 65% 2,2'-thiodiethanol (Sigma-Aldrich, St. Louis, MI, USA) (Cichewicz et al. 2017 [↗](#)) for DA staining. Images were acquired on a Nikon A1R confocal microscope with identical laser, filter and gain settings for all samples in each experiment. Immunofluorescence intensity levels were quantified using the Fiji software. Experiments were repeated independently at least 3 times on 4 to 6 brains per genotype.

Bang-sensitivity test

Bang-sensitivity assays were performed as previously described (Howlett et al. 2013 [↗](#)). 30-day-old males were divided into 5 groups of 10 flies under CO_2 and allowed to recover overnight. The following day, each group was placed in a vial without food and after 20 min of habituation, the vials were stimulated vigorously with a vortex mixer for 10 s at 2,500 rpm. The recovery time was measured for each fly, from the end of the stimulation until they reached a normal standing position. Results are the mean of the recovery time for at least 50 flies per genotype.

Drug administration

Allopurinol (A8003, Sigma Aldrich, St. Louis, MI, USA) was diluted in standard medium at 100 $\mu\text{g/ml}$ and flies were placed for 5 days on this medium before metabolite extraction. Adenine, adenosine and hypoxanthine (A2786, A9251 and H9377, Sigma Aldrich, St. Louis, MI, USA) or N^6 -

methyadenosine (m^6A) (QB-1055, Combi-Blocks, San Diego, CA, USA) were diluted in fly food medium at 500 μM . Parents were allowed to lay eggs on this medium in order to have exposition to the drug throughout all larval development of the progeny. Adults were collected and placed in normal medium until 5 days before the test, when they were placed again in food supplemented with adenosine or m^6A at the same concentrations.

Uric acid quantification

For purine metabolite extraction, 40 heads from 8-day-old flies were ground in 80% ethanol, heated for 3 min at 80°C and evaporated in a Speedvac apparatus. Dried residues were resuspended in MilliQ water and total protein content of each homogenate was measured with the Pierce BCA Protein Assay Kit (ThermoFisher Scientific, Waltham, MA, USA). 20 μl of each sample was injected into a 25 cm x 4.6 mm C18 Nucleosil column with 5- μm particles (Interchim, Montluçon, France). Purine metabolites were detected by an Agilent 1290 Infinity HPLC system coupled with a diode array detector as recommended by the ERDNIM advisory document. 7 wavelengths were used for detection (230, 240, 250, 260, 266, 270 and 280 nm) in order to have the spectrum of each compound for identification in addition to the retention time. The mobile phases contained 0.05 M monopotassium phosphate pH 5 and 65% (v/v) acetonitrile. The flux was fixed at 1 ml/min. For analysis, the maximum height of the compound was normalized to protein content and compared to the control genotype.

ATP assay

ATP level was measured by bioluminescence using the ATP Determination Kit (A22066, ThermoFisher Scientific, Waltham, MA, USA) on a TriStar 2 Spectrum LB942S microplate reader (Berthold Technologies, Bad Wildbad, Germany), according to the manufacturer's instructions. Samples were prepared as described previously (Fergestad et al. 2006 [\[1\]](#)). Briefly, 30 heads or 5 thoraces from 8-day-old flies were homogenized in 200 μL of 6 M guanidine-HCl to inhibit ATPases, boiled directly 5 min at 95°C and then centrifuged for 5 min at 13,000 g and 4°C. Total protein content of each supernatant was measured with the BCA Protein Assay Kit. Each supernatant was then diluted at 1:500 in TE buffer pH 8 and 10 μl were placed in a 96-well white-bottom plate. The luminescent reaction solution (containing D-luciferine, recombinant firefly luciferase, dithiothreitol and reaction buffer) was added to each well with an injector and high-gain 1 sec exposure glow reads were obtained at 15 min after reaction initiation. Results were compared to a standard curve generated with known ATP concentrations and final values were normalized to the protein content.

Enzyme activity assay

HGPRT and APRT activities were assessed in *Drosophila* by adapting methods previously established for human cells (Cartier and Hamet 1968 [\[2\]](#), Ceballos-Picot et al. 2009 [\[3\]](#)). 20 whole male flies were homogenized in 250 μl of 110 mM Tris, 10 mM $MgCl_2$ pH 7.4 (Tris- $MgCl_2$ buffer), immediately frozen and kept at least one night at $\pm 80^\circ C$ before the assay. After 5 min of centrifugation at 13,000 g, total protein content of each supernatant was measured with the BCA Protein Assay Kit to normalize activity level. Kinetics of [^{14}C]-hypoxanthine conversion to IMP (or in some cases [^{14}C]-guanine conversion to GMP), and [^{14}C]-adenine conversion to AMP were assessed for HGPRT and APRT assays, respectively. Compositions of reaction mediums were: 25 μl of a radioactive solution made with 38 μl of [^{14}C]-hypoxanthine (25 $\mu Ci/ml$) diluted in 1 ml of 1.2 mM cold hypoxanthine (or in some cases 38 μl of [^{14}C]-guanine (25 $\mu Ci/ml$) diluted in 1 ml of 1.2 mM cold guanine), for HGPRT assay, or 25 μl of a radioactive solution made with 75 μl of [^{14}C]-adenine (50 $\mu Ci/ml$) diluted in 1 ml of 1.2 mM cold adenine for APRT assay, and a volume of fly extract equivalent to 200 μg protein diluted in Tris- $MgCl_2$ buffer to 150 μl . Reactions were monitored at 37°C and started by adding 25 μl of the co-factor 5-phosphoribosyl-1-pyrophosphate (PRPP) at 10 mM. After 6, 12, 24 and 36 min, 40 μL of each pool was placed in a tube containing either 25 μl of HIE (3 mM hypoxanthine, 6 mM IMP, 200 mM EDTA) or AAE (3 mM adenine, 6 mM

AMP, 200 mM EDTA) solutions and incubated for 3 min at 95°C to stop the reaction. The different radioactive compounds were separated by paper chromatography on Whatman 3MM strips using 28% NH₄OH, 50 mM EDTA as solvent for about 1h30. Then, the substrates and products were visualized under a UV lamp at 254 nm and placed separately in vials in 2 ml of Scintran (VWR, Radnor, PA, USA). The radioactivity in disintegrations per minute (dpm) was measured in a Packard Tri-Carb 1600 TR liquid scintillation analyzer (PerkinElmer, Waltham, MA, USA). The percentage of substrate transformation as a function of time was converted in nmol/min/mg protein and finally normalized to wild-type control values.

Protein extraction and western blots

30 heads of 8-10 day-old-males per genotype were homogenized in 30 µL Laemmli buffer containing protease inhibitor (Complete mini Protease Inhibitor Cocktail, Roche Diagnostics), using a Minilys apparatus (Bertin Instruments, Montigny-le-Bretonneux, France). The lysates were incubated on ice for 30 min and centrifuged at 8,000 g for 10 min at 4°C. The extracted proteins were heated at 95°C for 10 min. Western blots were performed as previously described (Issa et al. 2018 [\[4\]](#)). Briefly, proteins were separated in 4–12% Novex NuPAGE Bis-Tris precast polyacrylamide gels (Life Technologies) following the manufacturer's protocol in a MOPS-SDS running buffer. A semi-dry transfer was done onto polyvinylidene difluoride membranes (Amersham Hybond P 0.45 µm) using a Hoefer TE77 apparatus. Membranes were probed overnight at 4°C with mouse monoclonal anti-TH (1:5000, cat. n° 22941, ImmunoStar, Hudson, WI, USA) and mouse monoclonal anti-actin beta (1:5000, cat. n° ab20272, Abcam, Cambridge, UK) that cross-reacts with *Drosophila* Actin 5C (Act5C). After incubation with horseradish peroxidase (HRP)-conjugated anti-mouse (1:5000, cat. n° 115–035–146, Jackson ImmunoResearch) as secondary antibody, immunolabeled bands were revealed by chemiluminescence staining using ECL RevelBLOT Intense (Ozyme, Saint-Cyr-l'École, France, cat. n° OZYB002-1000) and then digitally acquired with the ImageQuant TL software (GE Healthcare Life Science). Densitometry measures were made with the Fiji software and normalized to Act5C values as internal controls.

Reverse transcription-PCR and quantitative PCR

Total RNA was isolated by standard procedure from 20-30 heads of 8-day-old males collected on ice and lysed in 600 µL QIAzol Reagent (Qiagen, Venlo, Netherlands). 1 µg of total RNA was treated by DNase (DNase I, RNase-free, ThermoFisher Scientific, Waltham, MA, USA) according to the manufacturer's instructions. 5 µL of treated RNA was reverse transcribed using oligo d(T) primers (PrimeScript RT Reagent Kit, Takara Bio, Kyoto Japan). Then, at least 750 ng of the first strand cDNA was amplified in 20 µL of reaction mixture using PrimeStar Max DNA polymerase (Takara Bio, Kyoto Japan) with a Techne Prime Thermal Cycler apparatus (Bibby Scientific, Burlington, NJ, USA). The program cycles included 10 s denaturation at 98°C, 10 s annealing at 55°C and 10 s elongation at 72°C, repeated 30 times. PCR product levels were measured after electrophoresis by densitometry analysis with the Fiji software. Data were normalized to amplification level of the ribosomal protein *rp49/RpL32* transcripts as internal control. Sequences of the primers used were: sense *HPRT*, 5'-GAGATACAAAATGGCGACCCGACGCCCT; antisense *HPRT*, 5'-GCTCGGATCCTTATCATTACTAGGCTTTG (amplicon 686bp); sense *rp49*, 5'-GACGCTCAAGGGACAGTATC; antisense *rp49*, 5'-AAACGCGTTCTGCATGAG (amplicon 144bp).

For RT-qPCR, approximately 40 ng of the first strand cDNA was amplified in 10 µL of reaction mixture using the LightCycler 480 SYBR Green I Master reaction mix (Roche Applied Science, Mannheim, Germany) and a LightCycler 480 Instrument (Roche Applied Science, Mannheim, Germany). The program cycles included a 10-min preincubation step at 95°C, 40 cycles of amplification (10 s denaturation at 95°C, 10 s annealing at 55°C, 20 s elongation at 72°C), followed by a melting curves analysis for PCR product identification. Data were normalized to amplification level of the ribosomal protein *rp49/RpL32* transcripts as internal control. The genes analyzed and primer sequences used for qPCR are indicated in Table S4.

Adenosine assay

Adenosine was determined by Ultra Performance Liquid Chromatography (UPLC). 5 whole flies or 30 heads were homogenized in 120 μ l 0.9% (w/v) NaCl using a Minilys apparatus (Bertin Instruments, Montigny-le-Bretonneux, France) and frozen at $\mp 80^{\circ}\text{C}$. After unfreezing and 5 min of microcentrifugation, 20 μ l of 10% perchloric acid were added to 70 μ l of the supernatant and the mixture was left for 5 min on ice. After a new centrifugation, 20 μ l of a neutralization solution (made by mixing 3 volumes of 3 M K_2CO_3 with 1 volume of 6.4 mM NaOH containing 0.4 mg/ml bromothymol blue) were added and the mixture was centrifuged again before injection (5 μ l). Samples were analyzed with a UV diode-array detector on an Acquity UPLC HSS T3 column (1.8 μ m, 2.1 x 150mm) (Waters Corporation, Milford, MA, USA). The mobile phases consisted in Buffer A (30 mM ammonium acetate, pH 4.0 with 1:10,000 heptafluorobutyric acid (HPFA)) and Buffer B (acetonitrile with 1:10,000 HPFA), using a flow rate of 0.3 ml/min. Chromatographic conditions were: 3.5 min 100% Buffer A, 16.5 min up to 6.3% Buffer B, 2 min up to 100% Buffer B, and 1 min 100% Buffer B. The gradient was then returned over 5 min to 100% Buffer A, restoring the initial conditions. Results were normalized to protein levels for each sample.

Statistics

Statistical significance was determined with the Prism 6 software (GraphPad Software, La Jolla, CA, USA). Survival curves for longevity experiments were analyzed using the log-rank test. The Student's *t*-test was used to compare two genotypes or conditions, and one-way or two-way ANOVA with Tukey's, Dunnett's or Sidak's *post-hoc* multiple comparison tests for three or more conditions. Results are presented as mean $\pm$ SEM. Probability values in all figures: **p* < 0.05, ***p* < 0.01, ****p* < 0.001.

Acknowledgements

We are grateful to Pr. Hyder A. Jinnah for HPRT1 cDNA gift and Dr Thomas Riemensperger for helpful discussion. Part of the phylogenetic analyses were performed using the computing facilities of the PRABI-AMSB bioinformatics platform, Laboratory of Biometry and Evolutionary Biology, Lyon, France. This work was supported by fundings from CNRS, PSL Research University and ESPCI Paris to SB, Institut de France and Association Malaury to ICP. CP was recipient of PhD fellowships from Association Lesch-Nyhan Action (LNA), Association Malaury and Labex MemoLife.

Authors contributions

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Editors

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Reviewer #1 (Public Review):

The current manuscript focusses on the adenine phosphoribosyltransferase (Ap_{rt}) and how the lack of its function affects nervous system function. It puts it into the context of Lesch-Nyhan disease, a rare hereditary disease linked to hypoxanthine-guanine phosphoribosyltransferase (HGPRT). Since HGPRT appears absent in *Drosophila*, the study focusses initially on Ap_{rt} and shows that ap_{rt} mutants have a decreased life-span and altered uric acid levels (the latter can be attenuated by allopurinol treatment). Moreover, ap_{rt} mutants show defects in locomotor reactivity behaviors. A comparable phenotype can be observed when specifically knocking down ap_{rt} in dopaminergic cells (in an adult-specific fashion). Interestingly, also glia-specific knock-down caused a similar behavioral defect, which could not be restored when re-expressing UAS-ap_{rt}, while neuronal re-expression did restore the mutant phenotype. Moreover, mutants, pan-neuronal and glia-specific RNAi for ap_{rt} caused sleep-defects. Based on immunostainings Dopamine levels are increased; UPLC shows that adenosine levels are reduced and PCR showed an increase of Ent2 levels are increased (but not AdoR). Moreover, ap_{rt} mutants display seizure-like behaviors, which can be partly restored by purine feeding (adenosine and N6-methyladenosine). Finally, expression of the human HGPRT also causes locomotor defects.

The authors provide a wide range of genetic experimental data to assess behavior and some molecular assessment on how the defects may emerge. It is clearly written, and the arguments follow the experimental evidence that is provided.

The findings provide a new example of how manipulating specific genes in the fruit fly allow the study of fundamental molecular processes that are linked to a human disease.

<https://doi.org/10.7554/eLife.88510.2.sa2>

Reviewer #2 (Public Review):

The manuscript by Petitgas et al demonstrates that loss of function for the only enzyme responsible for the purine salvage pathway in fruit-flies reproduces the metabolic and neurologic phenotypes of human patients with Lesch-Nyhan disease (LND). LND is caused by mutations in the enzyme HGPRT, but this enzyme does not exist in fruit-flies, which instead only have Ap_{rt} for purine recycling. They demonstrate that mutants lacking the Ap_{rt} enzyme accumulate uric acid, which like in humans can be rescued by feeding flies allopurinol, and

have decreased longevity, locomotion and sleep impairments and seizures, with striking resemblance to HGPRT loss of function in humans. They demonstrate that both loss of function throughout development or specifically in the adult ubiquitously or in all neurons, or dopaminergic neurons, mushroom body neurons or glia, can reproduce the phenotypes (although knock-down in glia does not affect sleep). They show that the phenotypes can be rescued by over-expressing a wild-type form of the *Aprt* gene in neurons. They identify a decrease in adenosine levels as the cause underlying these phenotypes, as adenosine is a neurotransmitter functioning via the purinergic adenosine receptor in neurons. In fact, feeding flies throughout development and in the adult with either adenosine or m6A could prevent seizures. They also demonstrate that loss of adenosine caused a secondary up-regulation of ENT nucleoside transporters and of dopamine levels, that could explain the phenotypes of decreased sleep and hyperactivity and night. Finally, they provide the remarkable finding that over-expression of the human mutant HGPRT gene but not its wild-type form in neurons impaired locomotion and induced seizures. This means that the human mutant enzyme does not simply lack enzymatic activity, but it is toxic to neurons in some gain-of-function form. Altogether, these are very important and fundamental findings that convincingly demonstrate the establishment of a *Drosophila* model for the scientific community to investigate LND, to carry out drug testing screens and find cures.

The authors have dealt with my concerns satisfactorily and have explained the instances in which resolving experimentally the criticisms raised would require a work effort well beyond the scope of a revision for this manuscript.

<https://doi.org/10.7554/eLife.88510.2.sa1>

Reviewer #3 (Public Review):

The revised study provides better evidence to suggest that loss of *Aprt* activity in *Drosophila* provides a model for the loss of HGPRT activity in humans, which is causative for LND. Analysis of *Drosophila Aprt* mutations and RNAi-mediated knockdown reveals similar phenotypes to LND, particularly neurological defects, reduced nighttime sleep, and potentially seizures. LND is currently resistant to treatments and screening of a limited number of compounds in *Drosophila* has not identified a compound that can reduce all of the associated phenotypes. It is appropriate, therefore, that claims to have developed a clinically exploitable model for human LND have been toned down. Future drug screening may well prove profitable, but currently the evidence that *Drosophila Aprt* will be a suitable model for LND remains speculative.

The second approach adopted is to express a 'humanised mutated' form of HGPRT in *Drosophila*, which holds more promise for the development of a pharmacological screen. In particular, the locomotor defect is recapitulated but the seizure-like activity, whilst reported as being recapitulated, is debatable. A recovery time of 2.3 seconds is very much less than timings for typical seizure mutants. Nevertheless, the SING behaviour could be sufficient to screen against. However, this is not explored. With respect the short seizure duration, the authors cite similar findings for porin loss of function, but the cited study similarly did not employ anti-seizure drug exposure to validate that this phenotype is seizure related.

In summary, this is a largely descriptive study reporting the behavioural effects of an *Aprt* loss-of-function mutation. RNAi KD and rescue expression studies suggest that a mix of neuronal (particularly dopaminergic and possibly adenosinergic signalling pathways) and glia are involved in the behavioural phenotypes affecting locomotion, sleep and seizure. There remains insufficient evidence to have full confidence that the *Aprt* fly model will prove valuable for understanding / treating LND.

Author Response

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

The current manuscript focuses on the adenine phosphoribosyltransferase (Aprt) and how the lack of its function affects nervous system function. It puts it into the context of Lesch-Nyhan disease, a rare hereditary disease linked to hypoxanthine-guanine phosphoribosyltransferase (HGPRT). Since HGPRT appears absent in Drosophila, the study focuses initially on Aprt and shows that aprt mutants have a decreased life-span and altered uric acid levels (the latter can be attenuated by allopurinol treatment). Moreover, aprt mutants show defects in locomotor reactivity behaviors. A comparable phenotype can be observed when specifically knocking down aprt in dopaminergic cells. Interestingly, also glia-specific knock-down caused a similar behavioral defect, which could not be restored when re-expressing UAS-aprt, while neuronal re-expression did restore the mutant phenotype. Moreover, mutants, pan-neuronal and pan-neuronal plus glia RNAi for aprt caused sleep-defects. Based on immunostainings Dopamine levels are increased; UPLC shows that adenosine levels are reduced and PCR showed an increase of Ent2 levels are increased (but not AdoR). Moreover, aprt mutants display seizure-like behaviors, which can be partly restored by purine feeding (adenosine and N6methyladenosine). Finally, expression of the human HGPRT also causes locomotor defects.

The authors provide a wide range of genetic experimental data to assess behavior and some molecular assessment on how the defects may emerge. It is clearly written, and the arguments follow the experimental evidence that is provided. The findings provide a new example of how manipulating specific genes in the fruit fly allows the study of fundamental molecular processes that are linked to a human disease.

We thank the reviewer for his clear understanding and positive assessment of our work.

Reviewer #2 (Public Review):

The manuscript by Petitgas et al demonstrates that loss of function for the only enzyme responsible for the purine salvage pathway in fruit-flies reproduces the metabolic and neurologic phenotypes of human patients with Lesch-Nyhan disease (LND). LND is caused by mutations in the enzyme HGPRT, but this enzyme does not exist in fruit-flies, which instead only have Aprt for purine recycling. They demonstrate that mutants lacking the Aprt enzyme accumulate uric acid, which like in humans can be rescued by feeding flies allopurinol, and have decreased longevity, locomotion and sleep impairments and seizures, with striking resemblance to HGPRT loss of function in humans. They demonstrate that both loss of function throughout development or specifically in the adult ubiquitously or in all neurons, or dopaminergic neurons, mushroom body neurons or glia, can reproduce the phenotypes (although knock-down in glia does not affect sleep). They show that the phenotypes can be rescued by over-expressing a wild-type form of the Aprt gene in neurons. They identify a decrease in adenosine levels as the cause underlying these phenotypes, as adenosine is a neurotransmitter functioning via the purinergic adenosine receptor in neurons. In fact, feeding flies throughout development and in the adult with either adenosine or m6A could prevent seizures. They also demonstrate that loss of adenosine caused a secondary up-regulation of ENT nucleoside transporters and of dopamine levels, that could explain the phenotypes of

decreased sleep and hyperactivity and night. Finally, they provide the remarkable finding that over-expression of the human mutant HGPRT gene but not its wild-type form in neurons impaired locomotion and induced seizures. This means that the human mutant enzyme does not simply lack enzymatic activity, but it is toxic to neurons in some gain-of-function form. Altogether, these are very important and fundamental findings that convincingly demonstrate the establishment of a Drosophila model for the scientific community to investigate LND, to carry out drug testing screens and find cures.

We thank the reviewer for his clear understanding and positive assessment of our work.

The experiments are conducted with great rigour, using appropriate and exhaustive controls, and on the whole the evidence does convincingly or compellingly support the claims. The exception is an instance when authors mention 'data not shown' and here data should either be provided, or claims removed: "feeding flies with adenosine or m6A did not rescue the SING phenotype of Aprt mutants (data not shown)". It is important to show these data (see below).

As recommended by the reviewer, these results are now shown in the new Figure S15.

Sleep is used to refer to lack of movement of flies to cross a beam for more than 5 minutes. However, lack of movement does not necessarily mean the flies are asleep, as they could be un-motivated to move (which could reflect abnormal dopamine levels) or engaged in incessant grooming instead. These differences are important for future investigation into the neural circuits affect by LND.

We agree that the method we used could overestimate sleep duration because flies that don't move do not necessarily sleep either, as it is the case with brain-dopamine deficient flies (Riemensperger et al., PNAS 2011). To address this issue, we have recorded video data showing that after 5 min of inactivity, wild-type and Aprt5 mutant flies are less sensitive to stimulation, indicating that they were indeed asleep. This is now shown in the new Figure S10 and mentioned on page 17, lines 338-339 in the main text. In addition, in this work we report that Aprt mutant flies have a nocturnal insomnia phenotype. Sleep overestimation is not, therefore, an issue that could challenge these results.

The authors claim that based on BLAST genome searchers, there are no HPRTI (encoding HGPRT) homologues in Drosophila. However, such a claim would require instead structure-based searches that take into account structural conservation despite high sequence divergence, as this may not be detected by regular BLAST.

To reinforce our conclusions about the lack of homologue of the human HPRT1 gene in Drosophila, we have now added a Results section about the evolution of HGPRT proteins on pages 6-7, lines 122150, and two phylogenetic analyses as new Figures S2 and S3 with more details in legends. We have also carried out structural similarity searches against the RCSB PDB repository. The structural analysis did not identify any relevant similarity with HGPRT 3D structures in Insecta (mentioned lines 146-150). We hope these new analyses address the Reviewer's concerns. Furthermore, as shown in Table S2, no enzymatic HGPRT activity could be detected in extracts of wild-type Drosophila. A protein that would be structurally similar to human HGPRT but with a divergent sequence could not be involved in purine recycling without expressing HGPRT-like activity. In contrast, enzymatic Aprt activity could be easily detected in this organism (Figure S4 and Table S1).

This work raises important questions that still need resolving. For example, the link between uric acid accumulation, reduced adenosine levels, increased dopamine and behavioural neurologic consequences remain unresolved. It is important that they show

that restoring uric acid levels does not rescue locomotion nor seizure phenotypes, as this means that this is not the cause of the neurologic phenotypes.

We agree with the reviewer about the potential importance of our results and the need to resolve the exact origin of the neurological phenotypes. This would need to be addressed in further studies in our opinion. The fact that allopurinol treatment did not improve the locomotor ability of Aprt5 mutant flies is now shown in Figure 1D, E to emphasize this result. Results showing that allopurinol does not rescue the bang-sensitivity phenotype of Aprt-deficient mutants are shown in Figure S14.

Instead, their data indicate adenosine deficiency is the cause. However, one weakness is that for the manipulations they test some behaviours but not all. The authors could attempt to improve the link between mechanism and behaviour by testing whether over-expression of Aprt in neurons or glia, throughout development or in the adult, and feeding with adenosine and m6A can rescue each of the behavioural phenotypes handled: lifespan, SING, sleep and seizures. The authors could also attempt to knock-down dopamine levels concomitantly with feeding with adenosine or m6A to see if this rescues the phenotypes of SING and sleep.

The reviewer is right. However, carrying out all these experiments properly with enough repeats will require about two more years of work. Because of that, they could not be included in the revision of the present article. Here we show that Aprt overexpression in neurons, but not in glia, rescues the SING phenotype of Aprt5 mutants (Figure 2B and 2E). We have also added in the revised article the new result that Aprt overexpression reduces transcript levels of DTH1, which codes for the neural form of the dopamine-synthesizing enzyme tyrosine hydroxylase (new Figure 5F).

Visualising the neural circuits that express the adenosine receptor could reveal why the deficit in adenosine can affect distinct behaviours differentially, and which neurologic phenotypes are primary and which secondary consequences of the mutations. This would allow them to carry out epistasis analysis by knocking-down AdoR in specific circuits, whilst at the same time feeding Aprt mutants with Adenosine.

Deciphering the specific circuits involved in the various effects of adenosine would indeed be extremely interesting. Unfortunately very few is currently known about the neural circuits that express AdoR in flies. No antibody is available to detect this receptor in situ and mutated AdoR gene coding for a tagged form of the receptor has not been engineered yet to our knowledge.

The revelation that the mutant form of human HGPRT has toxic effects is very intriguing and important and it invites the community to investigate this further into the future.

To conclude, this is a fundamental piece of work that opens the opportunity for the broader scientific community to use Drosophila to investigate LND.

We sincerely thank the reviewer for his thoughtful and positive comments on our work.

Reviewer #3 (Public Review):

The study attempts to develop a Drosophila model for the human disease of LND. The issue here, and the main weakness of this study, is that Drosophila does not express the enzyme, HGPRT, which when mutated causes LND. The authors, instead, mutate the functionally-related Drosophila Aprt enzyme. However, it is unknown whether Aprt is also a structural homologue. Because of this, it will likely not be possible to identify pharmacological compounds that rescue HGPRT activity via a direct interaction (unless

modelling predicts high conservation of substrate binding pocket between the two enzymes, etc).

As stated in our Provisional Responses prior to revision of the Reviewed Preprint, the enzymes APRT and HGPRT are actually known to be functionally and structurally related. We apologize for not providing this information in the original submission. This point is now made clearer in the revised article on page 39, lines 785-792. Indeed, both human APRT and HGPRT belong to the type I PRTases family identified by a conserved phosphoribosyl pyrophosphate (PRPP) binding motif, which is used as a substrate to transfer phosphoribosyl to purines. This binding motif is only found in PRTases from the nucleotide synthesis and salvage pathways (see: Sinha and Smith (2001) Curr Opin Struct Biol 11(6):733-9, doi: 10.1016/s0959-440x(01)00274-3). The purine substrates adenine, hypoxanthine and guanine share the same chemical skeleton and APRT can bind hypoxanthine, indicating that APRT and HGPRT also share similarities in their substrate binding sites (Ozeir et al. (2019) J Biol Chem. 294(32):11980-11991, doi: 10.1074/jbc.RA119.009087). Moreover, Drosophila Appt and Human APRT are closely related as the amino acid sequences of APRT proteins have been highly conserved throughout evolution (see Figure S5B in our paper).

An additional weakness is that the study does not identify a molecule that may act as a lead compound for further development for treating LND. Rather, the various rescues reported are selective for only a subset of the disease-associated phenotypes. Thus, whilst informative, this first section of the study does not meet the study ambitions.

In this study, we identify adenosine and N6-methyladenosine as rescuers of the epileptic behavior in Appt mutant flies (shown in Figure 7E, F). Interestingly, the same molecules have been found to rescue the viability of fibroblasts and neural stem cells derived from iPSCs of LND patients, in which de novo purine synthesis was prevented (discussed on page 38, lines 747-753). This suggests that the Drosophila model reported here could help to identify new genetic targets and pharmacological compounds capable to rescue HGPRT mutations in humans.

The second approach adopted is to express a 'humanised mutated' form of HGPRT in Drosophila, which holds more promise for the development of a pharmacological screen. In particular, the locomotor defect is recapitulated but the seizure-like activity, whilst reported as being recapitulated, is debatable. A recovery time of 2.3 seconds is very much less than timings for typical seizure mutants. Nevertheless, the SING behaviour could be sufficient to screen against. However, this is not explored.

We agree with the reviewer that it would be very interesting to do a pharmacological screen in this second LND model. However, we did not have the possibility to carry out such a screen yet.

In summary, this is a largely descriptive study reporting the behavioural effects of an Appt loss-of-function mutation. RNAi KD and rescue expression studies suggest that a mix of neuronal (particularly dopaminergic and possibly adenosinergic signalling pathways) and glia are involved in the behavioural phenotypes affecting locomotion, sleep and seizure. There is insufficient evidence to have confidence that the Appt fly model will prove valuable for understanding / treating LND.

Here we report many common phenotypes between the Appt fly model and the symptoms of LND patients (reduced longevity, locomotor problems, sleep defects, overproduction of uric acid that is rescued by allopurinol treatment...). Moreover, APRT and HGPRT enzymes are both functional and structural homologues, as explained in our answers. We also found that the same drugs can rescue the seizure-like phenotype in Appt-deficient flies and the viability

of LND fibroblasts and neural stem cells, derived from iPSCs of LND patients, in which de novo purine synthesis is prevented (Figure 7E, F). In many respects, our results therefore suggest that Aprt mutant flies could be useful to better understand LND, and potentially to screen for new therapeutic compounds.

From the Reviewing Editor:

(1) How are the pathways of purine catabolism different between flies and mammals? How does the absence of HGPRT and presence of only AGPRT affect purine catabolism? When did HGPRT appear in evolution?

Purine catabolism is quite similar in flies and mammals, except for the lack of urate oxidase in primates, as described in Figure S1. We added words in the revised article about purine anabolism/catabolism pathways lines 123-126 (see below our detailed response to Reviewer 1's Recommendations). HGPRT is present in Bacteria, Archea and Eukaryota, and nearly all animal phyla. However, BLAST search indicates that HGPRT homologues cannot be found in most insect species, such as *Drosophila*. To reinforce our conclusions about the lack of homologue of the human HPRT1 gene in *Drosophila melanogaster*, we have now added a Results section about the evolution of HGPRT proteins on pages 6-7, lines 122-150, and two phylogenetic analyses as new Figures S2 and S3 with details in legends.

*In addition to BLAST a structural based modelling method should be used to establish the loss of HGPRT in *Drosophila*.*

In agreement with the phylogenetic analyses, we have confirmed that no HGPRT enzymatic activity can be detected in wild-type *Drosophila* extract (Table S2). To complete these observations, as recommended by reviewer #2, we have carried out 3D structure-based searches in the RCSB Protein Data Bank. This enabled us to compare human HGPRT with all currently available protein structures. We found no *Drosophila* protein with a divergent sequence showing relevant structural similarity to human HGPRT. In contrast, this search identified proteins similar to human HGPRT in many other species of Eukaryota, Archea and Bacteria. This is now mentioned on page 7, lines 146-150 in the revised article.

(2) Of the three biochemical changes reported the change in dopamine levels should be validated by other methods given the unreliable nature of IHC.

As recommended by Reviewer #1, we have added the results of new experiments carried out by RTqPCR and Western blotting, which confirm the effect of Aprt mutation on brain dopamine levels. In addition, we added the consistent result that Aprt overexpression reduces transcript levels of DTH1. The results are shown in the new panels E to H of Figure 5 and mentioned in the text on page 20, lines 385-389.

(3) As suggested by reviewer 2 it would be helpful to clearly identify which of the three biochemical changes (DA, uric acid, adenosine) are responsible for the numerous behaviours tested. This is important because it is relevant for developing any therapeutic strategy arising from this study.

We agree that it would be very interesting to decipher the relationship between the different behaviors observed in mutant flies and the biochemical changes (dopamine, uric acid or adenosine). However, this would require a large amount of new experiments and it would probably double the size of our paper, which already includes many original data. In our opinion, such a detailed study should logically be the purpose of another article.

(4) There is concern regarding the robustness of the seizure data. Reviewer 3 has suggestions on how to address this.

See our answers to Reviewer 3's recommendations below.

(5) Editorial corrections and changes suggested by reviewers 2 and 3 need to be addressed.

As indicated in our answers, we have taken into account and when possible addressed the corrections and changes suggested by the reviewers.

(6) It is recommended that the authors tone down the relevance of this model for LND, particularly in the abstract. The focus should be on stating what is actually delivered.

As recommended by the reviewing editor, and to take in account the reserved comments of reviewer #3, we have toned down our affirmation that our new fly models are relevant for LND in the last sentences of the Abstract and Discussion, and also added a question mark in the subtitle of the Discussion on line 777. As mentioned in our provisional responses to the Public Reviews, we would like to emphasize, however, that reviewers #1 and #2 expressed more confidence than reviewer #3 in the potential usefulness of our work. Reviewer #1 indeed stated that: "The findings provide a new example of how manipulating specific genes in the fruit fly allows the study of fundamental molecular processes that are linked to a human disease", and reviewer #2 further wrote: "Altogether, these are very important and fundamental findings that convincingly demonstrate the establishment of a *Drosophila* model for the scientific community to investigate LND, to carry out drug testing screens and find cures", and added: "To conclude, this is a fundamental piece of work that opens the opportunity for the broader scientific community to use *Drosophila* to investigate LND".

Reviewer #1 (Recommendations For The Authors):

- An important prerequisite for the current study is that there appears to be no HGPRT "activity" in *Drosophila*. It is initially stated that there was previously no "HGPRT activity observed" in two papers from the 70ies. It would be important to corroborate this notion and provide some background on the /catabolism pathways. How shared or divergent are these pathways between *Drosophila* and mammals?*

In agreement with the pioneering studies of Becker (1974a, b), we have confirmed in this work that no HGPRT enzymatic activity can be detected in wild-type *Drosophila* extracts, as mentioned in Results on page 6, lines 127-130 and reported in Table S2. Purine catabolism is quite similar in flies and mammals, except for the lack of urate oxidase in primates, as shown in Figure S1. All the enzymes involved in purine anabolism/catabolism or recycling in humans have been conserved in *Drosophila* and humans, with the notorious exception of HPRT1.

*If there is no HGPRT gene, but only the APRT ortholog, what would this mean for the metabolites? Our enzymatic assays on *Drosophila* extracts indicated that hypoxanthine and guanine cannot be recycled into IMP and GMP, respectively, contrary to adenine which can be converted into AMP in flies. In the absence of HGPRT activity, GMP and IMP could be produced by de novo purine synthesis, or, alternatively, synthesized from AMP, which can be converted into IMP by the enzyme AMPD, and then IMP can be converted into GMP by the enzymes IMPDH and GMPS. These metabolic pathways are depicted in Figure S1A.*

Is the lack of HGPRT specific for Drosophila, insects (generally in invertebrates)? I feel clarifying this would provide more insight into the motivation of the experimental approach.

As suggested by the Reviewer and the Reviewing Editor, we have addressed the evolution of HGPRT proteins more precisely in the revision. We have added a section on this subject in Results on pages 67, lines 122-150, and two phylogenetic analyses as Figures S2 and S3 with details in legends. A phylogenetic analysis was carried out a few years ago by Giorgio Matassi, who is now co-author of this paper. The most striking result was the great impact of horizontal gene transfer in the evolution of HGPRT in Insects (Figures S2 and S3). Our analysis of the phyletic distribution of HGPRT proteins revealed their striking rareness in Insecta, and in particular, their absence in Drosophilidae. The PSIBlast search detected however a significant hit in *Drosophila immigrans* (accession KAH8256851.1). Yet, this sequence is 100% identical to the HGPRT of the Gammaroteobacterium *Serratia marcescens*. Indeed, a phylogenetic analysis showed that *D. immigrans* HGPRT clusters with the *Serratia* genus (see Figure S3). This can be interpreted either a contamination of the sequenced sample, or as a very recent horizontal gene transfer event. The second scenario is more likely for the corresponding nucleotide sequences differ by 5 synonymous substitutions (out of 534 positions). A powerful approach to try to understand the "origin" of the *D. immigrans* protein would be to analyze whether horizontal gene transfer has affected its chromosomal neighbours. This approach, proposed previously by G. Matassi (BMC Evol Biol, 2017, 17:2, doi: 10.1186/s12862-016-0850-6), is highly demanding in terms of computing time and would require an ad hoc study. We hope that these new analyses address the Reviewer's concerns.

- *On the mechanistic side on how the behavioral defects may arise, the authors show that dopaminergic neurons (and glia cells) are involved. One interesting finding is that dopamine immunostainings suggest increased dopamine levels. However, immunostainings are notorious for artifacts and do not provide a strong quantitative assessment. I feel it would be helpful to have an alternative technique to corroborate this finding.*

We agree with the reviewer and we added the results of further confirmatory experiments in the four new panels E-H of Figure 5, showing that: 1) the transcript levels of DTH1 (encoding the neuronal isoform of the dopamine-synthesizing enzyme tyrosine hydroxylase in *Drosophila*) are increased in *Aprt5* mutants compared to wild-type flies (new Figure 5E), 2) consistent with this, DTH1 transcript levels were found in contrast to be decreased when *Aprt* was overexpressed ubiquitously in flies (new Figure 5F), 3) Western blot experiments showed that DTH1 protein levels are also increased in *Aprt5* mutant flies compared to controls (new Figure 5G-H).

Reviewer #2 (Recommendations For The Authors):

*As mentioned in the public review, the behavioural phenotypes of decreased lifespan, SING, sleep and seizures could be tested for all manipulations: feeding with allopurinol, adenosine and m6A, and combining this with knock-down dopamine levels in PAMs or MBs. This could help dissect the relationship between mutations in *Aprt* and behaviour.*

We thank the reviewer for these suggestions, and, indeed, we would have liked to do all these experiments. However, as mentioned in our responses to the Public Reviews, carrying out these experiments properly with sufficient repeats would require about two more years of work. We have already accumulated a large amount of data, so we have decided to publish our results at this stage in order to make our new fly models available to the scientific

community. We are giving careful and due consideration to these experimental proposals and we hope to continue our investigation on this topic in the future.

It would also be helpful to find out which neurons and glia express AdoR. Perhaps there are already tools available the authors could test or at least check with the scRNAseq Fly Atlas (public Scope database).

Following the reviewer's recommendation, we have checked the scRNAseq Fly Atlas for AdoR expression in the brain, compared to that of ple (encoding tyrosine hydroxylase) and Eaaf1 (encoding the astrocytic glutamate transporter). As shown in the image below, the results are not very informative. AdoR appears to be expressed in rather widespread subsets of neurons and glial cells, that partly overlap with ple and Eaaf1 expression. Further work would be required to identify more precisely the neurons and glial cells expressing AdoR in the brain.

Author response image 1.



Page 7, line 161: use of the word 'normalize'. "We tried to normalise uric acid content in flies..." would best to use 'rescue' instead, as normalisation in science has a different meaning.

We modified this word as suggested.

Page 9 line 203: 'genomic deficiencies that cover': the genetic term is 'uncover', as a deficiency for a locus reveals a phenotypes, thus it is said 'a gene uncovered by xx deficiency'.

Thank you for this helpful remark. We corrected this in line 221.

Page 10, lines 206-208: 'allopurinol treatment did not improve the locomotor activity...'. These are important observations that should be best presented within the main manuscript Figure 1.

As recommended, we have transferred the graphs of Figure S5 to new panels D and E of Figure 1.

Figure 4: please indicate genotypes in the figure, where no information is given that these are UASAp^{rt}-RNAi experiments.

We added the complete genotype in Figure 4G, and also in Figure S12C and D. Thank you for noting that.

Page 25 line 491: "None of these drugs was able to rescue the SING defects (data not shown)". Either provide the data or remove this claim.

We have added these data in the new Figure S15.

Statistical analyses: details are provided in the methods, but the name of test and multiple comparisons corrections should be also provided in the legends.

Thank you very much for the careful proofreading. This was an oversight and we have added the information in all legends of the revised article.

Reviewer #3 (Recommendations For The Authors):

This is a difficult manuscript to appreciate. The abstract and introduction suggest that the study is to identify novel treatments for a human disease (LND) by development of a Drosophila model. Much of the results, however, are focussed to describing the consequences to purine metabolism of the Ap^{rt} mutation. To my mind, a rewrite to focus on the latter would be beneficial. The potential applicability to LND would be best restricted to the discussion.

We apologize for not making our goals clearer. Our purpose was to find out if purine recycling deficiency could lead to metabolic and neurobehavioral disturbances in Drosophila, as it is the case in human LND patients when HGPRT is mutated. Interestingly, we observed that mutation of the only purine recycling enzyme in flies, Ap^{rt}, did induce defects in part comparable to that of LND in humans, including overproduction of uric acid that is rescued by allopurinol treatment, reduced longevity, and various neurobehavioral phenotypes including bang-sensitive seizure, sleep defects and locomotor impairments. We also identified adenosine and N6-methyladenosine as rescuers of the epileptic behavior in these mutants. These drugs were also identified as therapeutic candidates in screens based on iPSCs from LND patients. This suggests that Ap^{rt} deficiency in Drosophila could be used as a model to better understand this disease and find new therapeutic targets.

Regardless of the above comment, the concluding sentence of the abstract is inappropriate. This study does not show that Drosophila can be used to identify a cure for LND.

We agree with the Reviewer that the last sentence of the abstract was too affirmative. As also recommended by the reviewing editor, we have modified this sentence in the abstract and other sentences in the text in order to tone down the affirmation that our new fly models are relevant for LND. See our answers to the Reviewing Editor above for details.

Indeed, I would challenge the premise that screening against a functional, but unknown if structural, homologue (Ap^{rt}) will ever provide an exploitable opportunity. To meet this statement, this study needs to identify a treatment that rescues all of the behavioural

phenotypes associated with the Aprt mutation, in addition to rescuing the influences of the mis-expression of mutated HGPRT.

APRT and HGPRT are both functionally and structurally related. Both human APRT and HGPRT belong to the type I PRTases family identified by a conserved phosphoribosyl pyrophosphate (PRPP) binding motif, which is used as a substrate to transfer phosphoribosyl to purines. This binding motif is only found in PRTases from the nucleotide synthesis and salvage pathways (see: Sinha and Smith (2001) Curr Opin Struct Biol 11(6):733-9733-9, doi: 10.1016/s0959-440x(01)00274-3). The purine substrates adenine, hypoxanthine and guanine share the same chemical skeleton and APRT can bind hypoxanthine, indicating that APRT and HGPRT also share similarities in their substrate binding sites (Ozeir et al. (2019) J Biol Chem. 294(32): 11980-11991, doi: 10.1074/jbc.RA119.009087)). This point has been made clearer in the Discussion page 39, in lines 785-792.. Finally, Drosophila Aprt and Human APRT are closely related as the amino acid sequences of APRTs have been highly conserved throughout evolution (shown in Figure S5B).

With respect to expression of the mutated HGPRT: the short seizure recovery time of 2.3 seconds is not very convincing evidence of a seizure phenotype. This is far below the timings reported for typical BS mutations. Because of this, the authors should run a positive control (e.g. one of the well-established BS mutations: parabss, eas or jus) to validate their assay. Moreover, was the seizure induced by the Aprt mutation (17.3 secs - again a low value) rescued by prior exposure to an antiepileptic? Could this behaviour be, instead, related to the SING locomotor phenotype?

The assay we used to test for bang-sensitivity has been validated in previous articles from different laboratories. We agree that the recovery times we observed were shorter than those of the BS mutations mentioned by the reviewer. However, we could cite another Drosophila BS mutant, porin, that shows similarly short recovery times (2.5 and 6 sec, according to the porin alleles tested, Graham et al. J Biol Chem. 2010, doi: 10.1074/jbc.M109.080317). This is now mentioned on page 36 lines 717-720). In addition, the BS phenotype we observed with Aprt mutants was robust and highly significant compared to control flies (Figure 7). We did not try to rescue this phenotype by exposing the flies to an antiepileptic, but we do not think that it can be related to the SING phenotype. Indeed, providing adenosine or N6-methyladenosine to Aprt5 mutant flies was able to rescue the BS phenotype (Figure 7E, F), but did not rescue the locomotor defects (new Figure S15). Moreover, SING performances of Aprt5 mutant flies at 8 or 30 d a. E. are decreased nearly in almost identical way (Figure 1C), while we observed an effect on BS behavior at 30 d a. E., which implies that the SING and BS behaviors are most likely unrelated.

Line 731 states that 'Aprt mutants show a typical BS phenotype' - whilst accurate to some extent (e.g. the behaviour depicted in the supp videos), it should be made clear, it should be made clear that the recovery time is uncharacteristically short and thus differs from typical BS mutations.

We have corrected the sentence in the revised article to mention that (page 36, lines 717-718).

Line 732 stating that BS phenotype is often linked to neuronal activity - what other links would there be? Even if via glia or other tissues the final effect is via neurons.

We have modified this sentence (page 36, line 720).

The introduction and, particularly, the discussion are overly long and, in the case of the latter, repetitive of the results text. Pruning to make the paper more concise would be very beneficial. Removal of the extensive speculation about how DA and adenosine may

interact would help in this regard (line 688 onwards). Indeed, in many places the discussion morphs into a review.

We agree with the reviewer on this point, and have therefore done our best to shorten the Introduction and Discussion, which are now 24% and 21% shorter, respectively, in the revised article compared to the original submission.

The applicability of using Drosophila Aprt mutations to screen for compounds that may treat LND is predicated on some degree of similarity in either enzyme structure or metabolic pathways. A discussion of how relevant, therefore, studying Aprt is needs to be included. Given the authors insights - where should potential new rugs be targeted to?

As stated above, we now mention in the article that APRT and HGPRT share similarities in their structure. In addition, the metabolic pathways between humans and Drosophila have been largely conserved (shown in Figure S1B).